CLASS X GEOGRAPHY — EXAM REVISION NOTES

All 12 chapters · formulas & worked calculations · diagrams · model answers · past-paper pointers

Built from the **BCSE 2025** and **2023** Geography papers. Every question in both papers is solved inside.

How to use: Orange tags show where a topic appeared: P25 Q2a = 2025 paper question 2(a); P23 = 2023 paper. "Sec A" = Question 1 (MCQ / true-false / fill-in / match / map work). Green boxes are **solved questions** with the marks in brackets and key answers in **red**. Purple tag diagram = a picture or graph in the paper. **Chapters:** 1 Universe & solar system · 2 Earth, plates & geological time · 3 Landforms: wind, rivers & karst · 4 Climate, vegetation & houses · 5 Latitude, longitude & time · 6 Map reading & survey tools · 7 Outline-map work · 8 Population & settlements · 9 Agriculture & food · 10 Minerals, industry & energy · 11 Biodiversity, environment & tourism · 12 Natural disasters.

⚠ **Read this first — limits of these notes** (a) The **topographic map extracts** (Trashigang 78M/11 in 2023, Mongar 78M/3 in 2025) and the **photographs** were not in the PDFs, so answers that depend on them (spot heights, grid references, what is in grid square 2030, etc.) show the **method + the likely answer** and are tagged **(map)** — check them on your map sheet. (b) I had no textbook or marking scheme, so answers I could only reason out are tagged **(check)**: 2023 match-i (organic vs inorganic minerals), 2025 T/F-ii (solar energy), 2025 fill-in iii (zone name for 200–2000 m), the Pele La and Drangme Chhu map answers. (c) Time answers use Bhutan = GMT +6 (90°E). Prefer the wording of **your own textbook** and teacher.

Paper pattern (2023 & 2025)

Part	Content	Marks
Sec A Q1(a)	10 MCQs	10
Q1 (b)–(d)	true/false, fill-in, match (5 each; order differs by year)	15
Q1(e)	topographic map extract: 8–9 short parts (contour interval, grid reference, scale, drainage, suitability, tourism…)	15
Q1(f)	outline map of **Asia** (4 items)	4
Q1(g)	outline map of **Bhutan** (6 items)	6
Sec B Q2–Q7	attempt **ANY FIVE** of six; each question has 5 parts (a–e) worth 2 marks (some 1 or 3)	50

How to answer

- **One point = one mark.** "TWO reasons" [2] → two separate, explained points. Extra points earn nothing.
- **Agree / Do you think… Justify:** state *Yes* or *No* first, then the reasons asked. Link each reason to Bhutan.
- **Calculations:** write the formula, substitute, give the unit (e.g. 4 min per degree).
- **Map work:** use a pencil, thin neat lines, names in small capitals; use the symbol the question asks for (thick line, dot, ∧∧∧,][, DZ, OP).
- Do not write your name or school. Circle ONE letter only.

Topic map: where each chapter appeared

Chapter	2023	2025
1 Universe	Q1a(i),d(iii),Q2d	Q1a(iii),b(iii),c(i),Q5b
2 Earth & plates	Q1a(iv,v),d(iv),Q2e,Q5a,e	Q1a(vi),c(iv),Q2c,Q5a
3 Landforms	Q1a(vi),Q6a,b,Q7e	Q1d,Q7a,b
4 Climate etc.	Q1a(iii),Q3b,Q5c,d,Q7d	Q1a(ii,v),c(ii,iii),Q2e,Q7d
5 Time	Q1a(ii),c(i),Q3a,Q5b,Q7c	Q1a(iv),Q3e,Q4a,c
6 Map reading	Q1e(i–viii)	Q1a(i),b(i),e(i–ix),Q2a,Q5c
7 Outline maps	Q1f,g	Q1f,g
8 Population	Q1a(viii,ix),c(iii,v),Q3d,Q4a,b,Q7a	Q1a(x),Q2b,Q3a,c,Q4b,e,Q6c
9 Agriculture	Q1a(vii),d(v),Q3c,Q6d	Q2e,Q3b,Q4d,Q5e
10 Minerals/industry	Q1b,d(i),Q2a,b,Q4c	Q1a(viii),b(ii),Q2d,Q5d,Q6a,Q7c
11 Biodiversity	Q1c(ii),Q3e,Q6c,e	Q1a(vii),b(v),Q6d,Q7b,e
12 Disasters	Q1a(x),d(ii),Q2c,Q4d,Q7b	Q1a(ix),c(v),Q3d,Q6b,e

Repeated every year: Big Bang / solar nebula · plate tectonics & Himalayas · Wegener/Pangaea consequences · wind & karst landforms · time-zone calculation · RF → statement scale · Central Place Theory · population trends · mining & biodiversity "agree/justify" · earthquake / GLOF · Asia & Bhutan outline map.

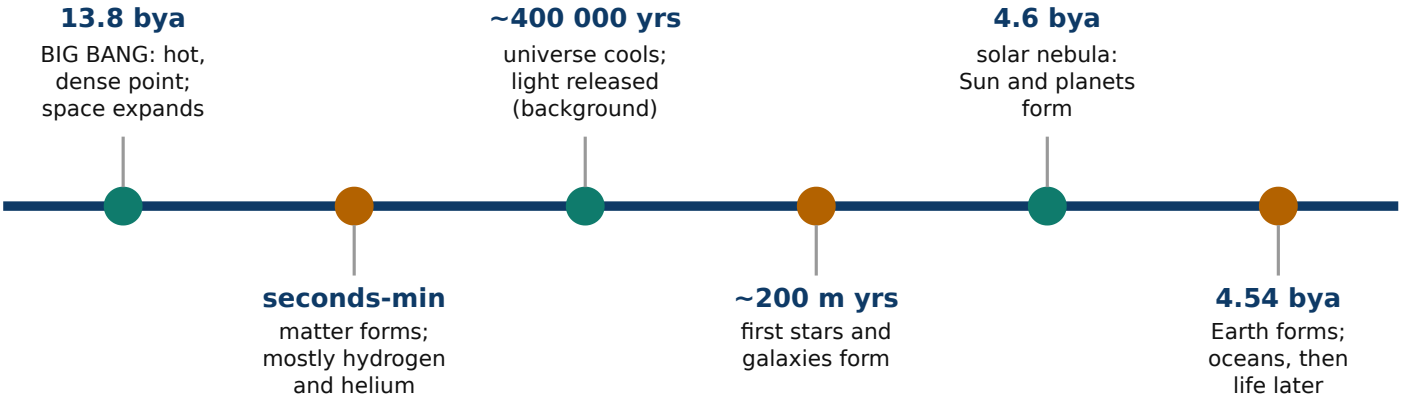
1 · THE UNIVERSE & THE SOLAR SYSTEM

P25 Q1a(iii),b(iii),c(i),Q5b · P23 Q1a(i),d(iii),Q2d,Q4e

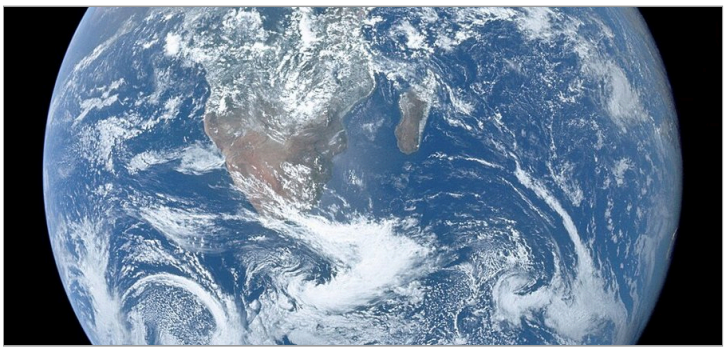
Key ideas

- **Universe** = all space, matter and energy. **Galaxy** = billions of stars held by gravity (our **Milky Way**). **Light-year** = distance light travels in one year (≈ 9.46 trillion km).
- **Big Bang Theory:** about **13.8 billion years ago** the universe began as a **single, extremely hot, dense point**, then **expanded** and cooled; matter formed stars and galaxies. Put forward by the Belgian priest **Georges Lemaître** (1927) P25 Q1a(iii): A; **Edwin Hubble** (1929) showed galaxies are moving apart (not the theorist of the Big Bang).
- **Expansion of the universe means a wider gap between galaxies** (space itself stretches; galaxies are not "new") P23 Q1a(i): B .
- **Evidence:** (1) **red shift**; (2) **cosmic microwave background** radiation (afterglow); (3) proportion of hydrogen (~75%) and helium (~25%).
- **Red shift** = light from distant galaxies is **stretched to longer (redder) wavelengths** because they are **moving away**; the farther the galaxy, the bigger the shift → the **universe is expanding** → it must have been smaller and denser in the past. P23 Q2d
- **Solar Nebula Theory** (Kant, Laplace): the solar system formed ~**4.6 billion years ago** from a rotating cloud of dust and gas (the **nebula**) P23 Q1d(iii): nebula. Gravity pulled the cloud in; it spun faster, flattened into a **disc**; the centre became the **Sun**.
- **Accretion:** dust grains stuck together → **planetesimals** → they collided and merged into larger **protoplanets** → planets. The process is called **accretion** P25 Q1c(i). The Big Bang does **not** explain planets forming from dust and gas → 2025 b(iii) **FALSE**.
- **Planets:** inner **terrestrial** (Mercury, Venus, Earth, Mars) – small, rocky, because heat drove light gases away; outer **giants** (Jupiter, Saturn gas giants; Uranus, Neptune ice giants).

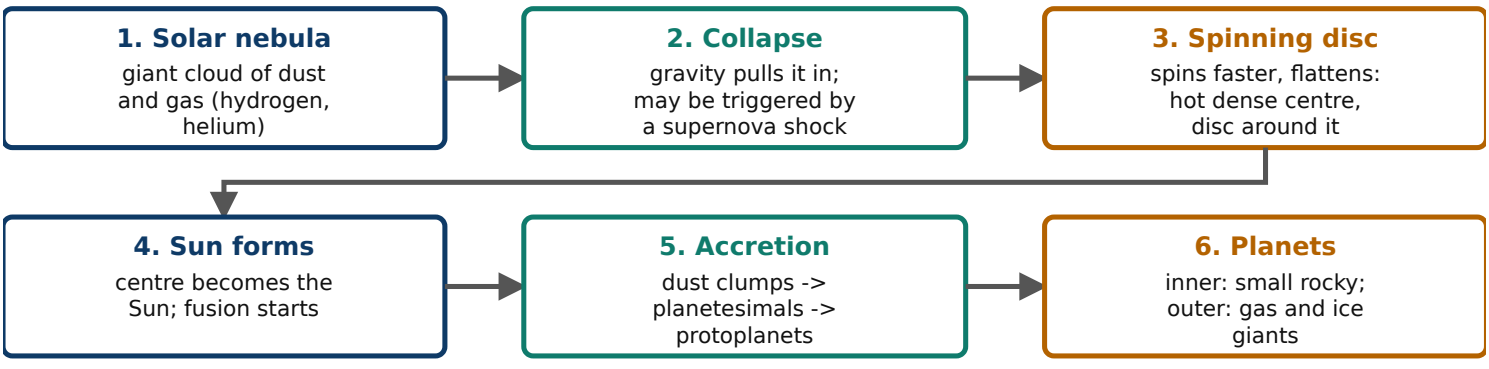
Timeline of the universe (bya = billion years ago)



From the Big Bang to the Earth.



Earth from space (Apollo 17, 1972) — the "Blue Marble".
Photo: NASA/Apollo 17 crew; taken by either Har..., Public domain, Wikimedia Commons.



Solar Nebula Theory: from cloud to planets.



The Orion Nebula (Hubble) — a cloud of gas and dust where new stars and planets form.
Photo: NASA, ESA, M. Robberto (Space Telescope..., Public domain, Wikimedia Commons.

Big Bang vs Solar Nebula

	Big Bang	Solar Nebula
Explains	origin and expansion of the universe	origin of the Sun and planets
Age	13.8 billion years	4.6 billion years
Key terms	expansion, red shift, CMB	nebula, disc, accretion

Solved questions

P25 Q5(b) [3] Why is the **Solar Nebula Theory** widely accepted? (1) It **explains why all planets orbit the Sun in the same direction and almost the same plane** (they formed from one spinning disc). (2) It explains the **two kinds of planets** — small rocky ones near the hot Sun, large gas/ice giants in the cold outer region. (3) **Telescopes (and space probes) observe young stars surrounded by discs of dust and gas**, and the theory fits the age of meteorites and Moon rocks (~4.6 billion years).

P23 Q2(d) [2] Red shift: **meaning** — the shift of light from a receding galaxy to **longer (red) wavelengths**, like the drop in pitch of a siren moving away. **It explains** that galaxies are moving away from each other, so the **universe is expanding** — evidence that it began from a single point (Big Bang).

P23 Q4(e) [2] Should Bhutan launch missions to planets like Mars? **No, not at present.** (1) Space missions are **extremely expensive** and need advanced technology and scientists that a small country does not have; (2) Bhutan's money is better used for **urgent needs** — health, education, climate and disaster safety. (Bhutan can instead **join international space cooperation**, such as small satellites with partners. "Yes" is accepted if justified, e.g. inspiring youth and developing technology.)

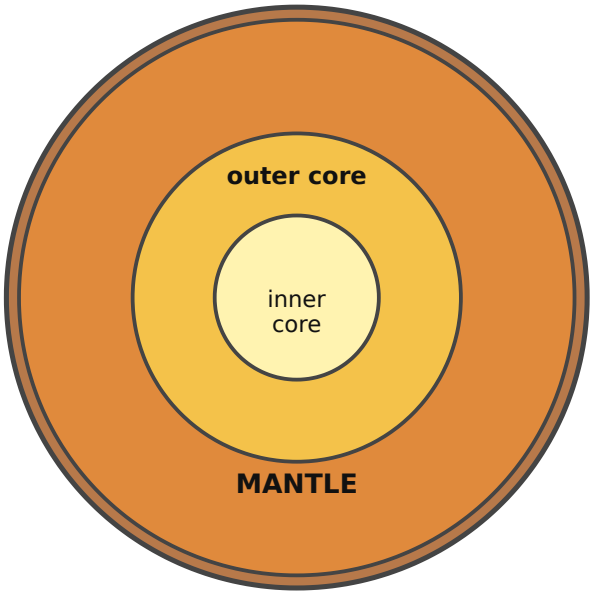
Quick answers: P23 Q1a(i) **B** · d(iii) **nebula**. P25 Q1a(iii) **A** (Lemaître) · b(iii) **FALSE** · c(i) **accretion**.

2 · EARTH, PLATE TECTONICS & GEOLOGICAL TIME

P25 Q1a(vi),c(iv),Q2c,Q5a · P23 Q1a(iv,v),d(iv),Q2e,Q5a,e

Structure of the Earth

- **Crust** (thin, solid; continental thicker, oceanic thinner) → **mantle** (~2,900 km, hot flowing rock) → **outer core** (liquid iron-nickel) → **inner core** (solid).
- **Lithosphere** = crust + rigid top of mantle, broken into **plates**. These plates "look like wooden blocks floating over a weak, ductile region", the **asthenosphere** P23 Q1a(v): D. **Tuzo Wilson** (1960s) gave the idea of plates of continental and oceanic crust.



CRUST

5-10 km (oceanic), 30-70 km (continental); solid rock

MANTLE

about 2,900 km thick; hot, slowly flowing rock

OUTER CORE

about 2,200 km; liquid iron and nickel (makes the magnetic field)

INNER CORE

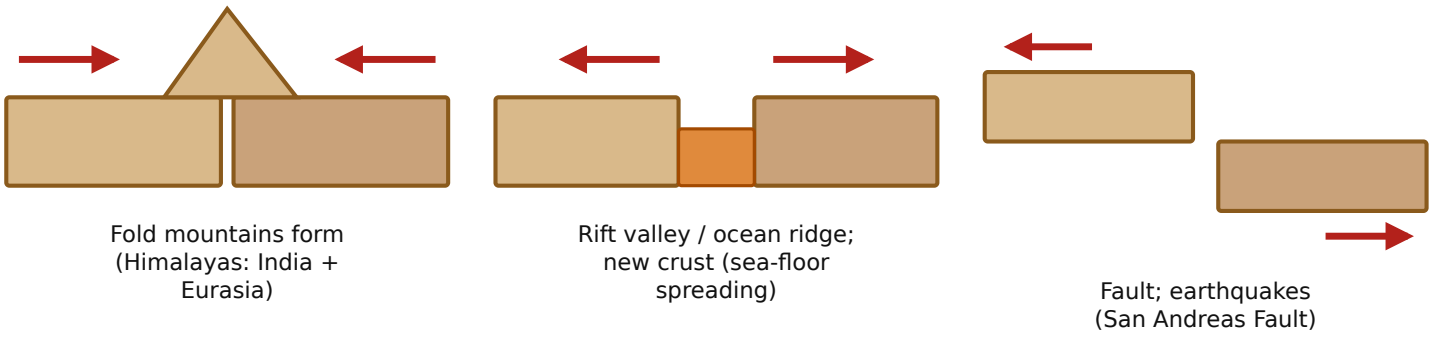
about 1,200 km; solid iron-nickel, ~5,500 C

LITHOSPHERE = crust + rigid top of mantle (~100 km)
floats on the ASTHENOSPHERE (weak, ductile layer)

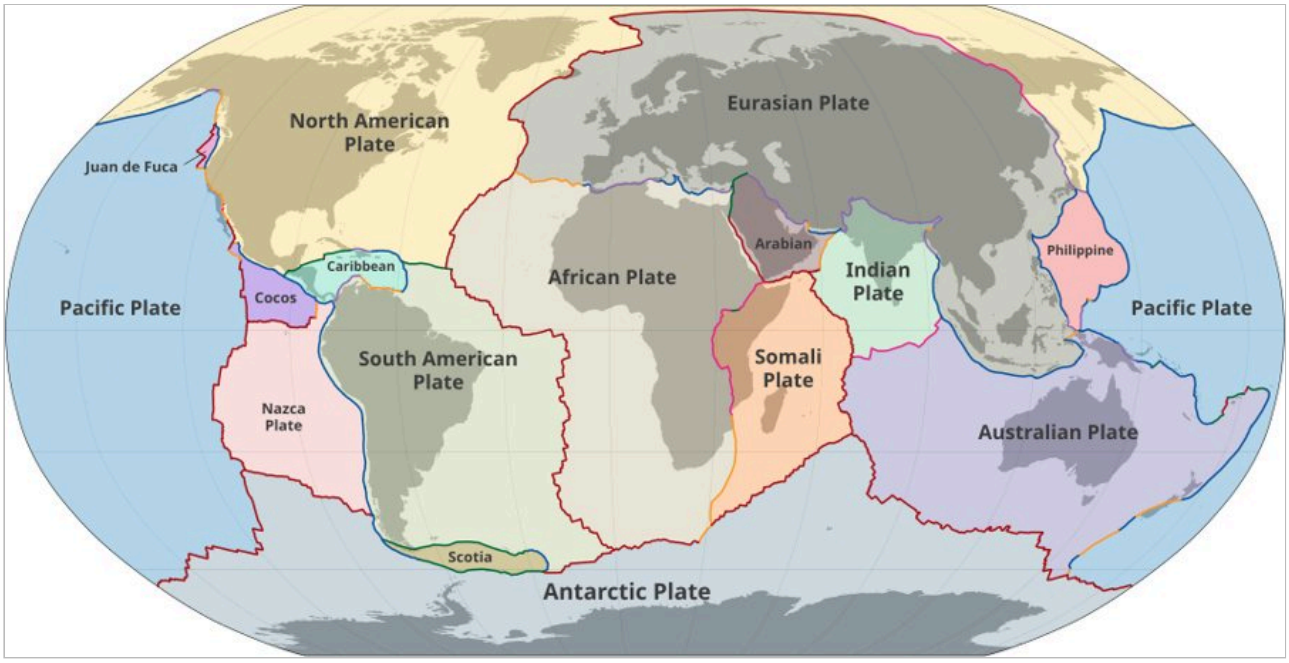
Layers of the Earth.

Continental drift & plate tectonics

- **Alfred Wegener (1912):** all continents were once joined as the supercontinent **Pangaea** ("all land") ~**200–300 million years ago**, which split and drifted into present positions P25 Q1c(iv): Pangaea. Pangaea first split into **Laurasia** (north) and **Gondwana** (south).
- **Evidence:** jigsaw fit of coasts; same fossils on separate continents; matching rock layers/mountains; glacial deposits in warm lands.
- **Plate boundaries: convergent** (collide: fold mountains, trenches, volcanoes, earthquakes), **divergent** (move apart: rift valleys, mid-ocean ridges), **transform** (slide past: earthquakes).
- **Himalayas:** the **Indian plate moved north and converged with the Eurasian plate** 50–70 million years ago; the sediments of the Tethys Sea between them were **crumpled and folded upward**, forming young **fold mountains** P23 Q1a(iv): C P25 Q1a(vi): B P23 Q1d(iv): fold mountains. Convergent movement presses both sides of the boundary and folds the rock layers. The plates are still pushing, so the Himalayas keep rising and Bhutan has **earthquakes**.
- **Proof the Himalayas were under the sea:** (1) **marine fossils** (ammonites, shells) found in Himalayan rocks; (2) **sedimentary rocks such as limestone and shale** formed on the sea bed lie at high altitude; (3) salt deposits and the sea-floor rocks (ophiolites) found there. P23 Q2e

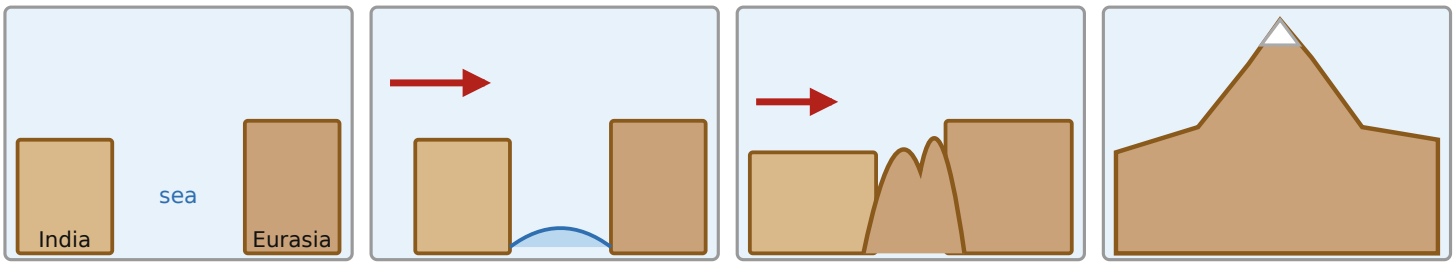


Three types of plate boundary.



Map: the major tectonic plates of the world.
Photo: M.Bitton, CC BY-SA 3.0, Wikimedia Commons.

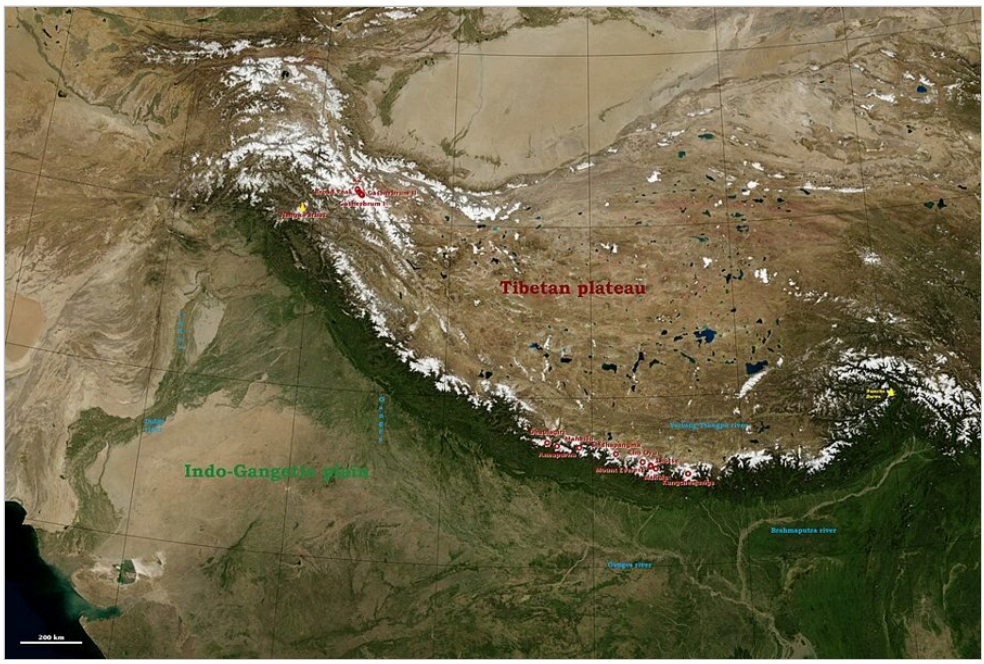
1. Tethys Sea | **2. India drifts north** | **3. Collision 50-70 mya** | **4. Himalayas today**



India (south) and Eurasia (north) | sea sediments squeezed | crumpling, folding | still rising ~ 4 mm/yr

Marine fossils (ammonites, shells) and marine limestone high in the Himalayas prove the area was once under the Tethys Sea.

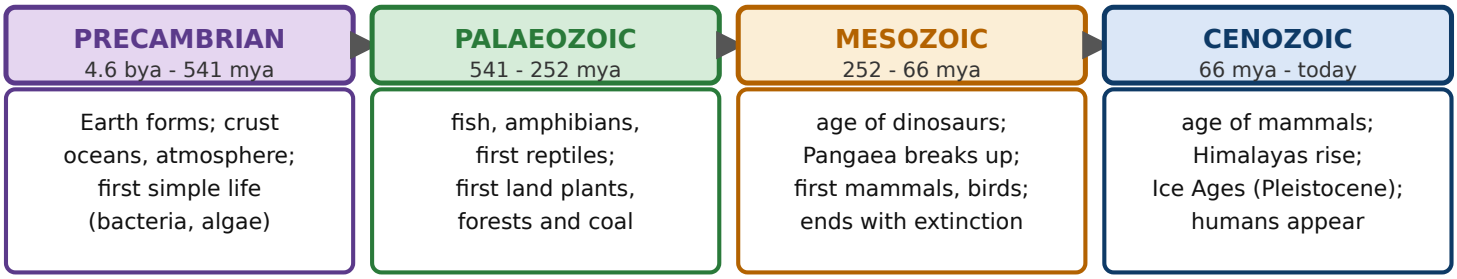
How the Himalayas formed.



Satellite map: the Himalayas, Tibetan plateau and Indo-Gangetic plain.
Photo: NASA Landsat image, Public domain, Wikimedia Commons.

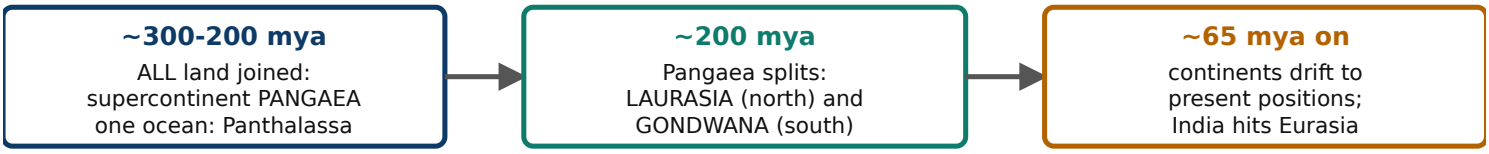
Geological Time Scale (GTS)

- A chronological system dating Earth's rock strata and history. Order from oldest: **Precambrian → Palaeozoic → Mesozoic → Cenozoic** P25 Q5a.



Order (oldest to newest): Precambrian -> Palaeozoic -> Mesozoic -> Cenozoic (i, iii, ii, iv in 2025 Q5a)

The four eras.



Break-up of Pangaea.

If all continents joined again (consequences)

1 Vast dry interior far from the sea: huge deserts, extreme hot/cold (continental climate)
2 One giant ocean, shorter coastline: less coastal rainfall, fewer marine habitats
3 Species mix; competition; extinction of some
4 Old mountains erode, new ones where plates collide

Evidence for continental drift (Wegener)

1 Jigsaw fit: S. America / Africa coasts
2 Same fossils (Mesosaurus, Glossopteris)
3 Matching rock strata and coalfields
4 Glacial deposits in now-warm lands
5 Later: sea-floor spreading, GPS plate movement

Consequences of one supercontinent, and the evidence for drift.

Solved questions

P25 Q5(a) [2] Chronological order: **i. Precambrian → iii. Palaeozoic → ii. Mesozoic → iv. Cenozoic** (i, iii, ii, iv).

P23 Q5(a) [2] FOUR events of the **Cenozoic era** (66 mya to today): (1) the **Himalayas and Alps rose** as the Indian plate collided with Eurasia; (2) **mammals spread** and became the dominant animals (Age of Mammals); (3) **Ice Ages** (Pleistocene) covered large areas and sea level rose and fell; (4) **modern humans appeared**; also the continents reached their present positions and grasslands spread.

P25 Q2(c) / P23 Q5(e) [2] If the continents stayed as one land mass (Pangaea) — TWO consequences: (1) **very dry continental interior** far from the sea: vast deserts and extreme hot summers/cold winters; (2) **one huge ocean, shorter coastline** → less coastal land, fewer marine habitats and less rainfall; also species would mix and some become extinct, and there would be no Himalayas or Indian Ocean as now. (Any two.)

P23 Q2(e) [2] The Himalayas were once under the sea — TWO evidences: (1) **marine fossils** such as ammonites and shells are found in the high rocks; (2) the rocks are **marine sedimentary rocks** (limestone, sandstone) deposited on the Tethys sea bed and later lifted when India collided with Eurasia.

Quick: P23 iv **C**, v **D**; d(iv) **fold mountains**. P25 vi **B**; c(iv) **Pangaea**.

3 · LANDFORMS: WIND, RIVERS & UNDERGROUND WATER (KARST)

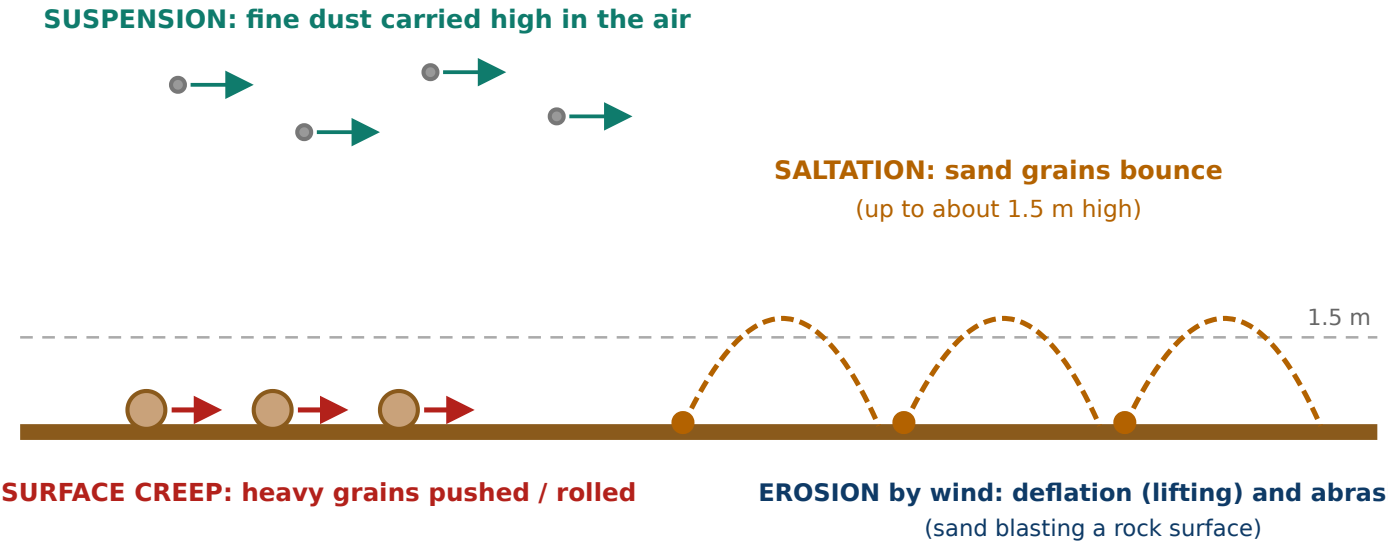
P25 Q1d,Q7a,b · P23 Q1a(vi),Q6a,b,Q7e

Agents of gradation

- **Erosion** (wearing away) → **transportation** (carrying) → **deposition** (dropping when energy is lost). Agents: running water, wind, glaciers, underground water, waves.
- **Deposition** happens when the river **loses energy and drops the material it was carrying** P25 Q1d(ii): a — e.g. delta, floodplain, alluvial fan.

Work of wind (arid areas)

- **Erosion: deflation** = wind lifts and blows away loose dust and sand P25 Q1d: f; **abrasion** = wind-carried particles strike and wear the rock surface (sand-blasting) P25 Q1d(iv): e.
- **Transportation** P25 Q1d: **suspension** — fine soil particles lifted and held in the air (v: **b**); **saltation** — grains bounce along the ground up to about 1.5 m (iii: **c**); **surface creep** — larger, heavy particles pushed and rolled along the ground (i: **d**).
- **Erosional features: mushroom (pedestal) rocks**, **yardangs** and zeugen (ridges), deflation hollows/ oases, desert pavement.
- **Depositional features: sand dunes** (barchan = crescent-shaped; seif = long), **loess** (fine wind-blown silt, fertile).



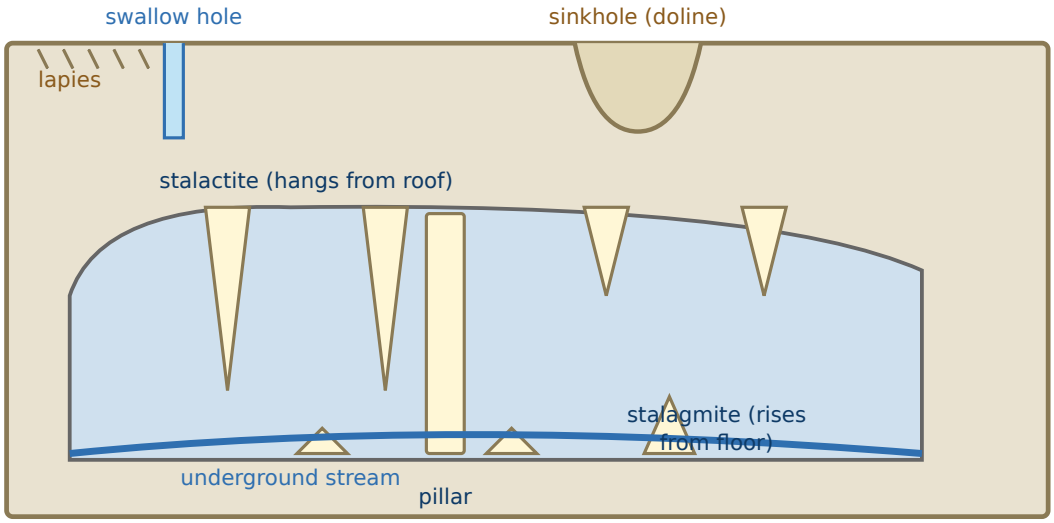
How wind moves material.



Sand dunes — a depositional landform of the wind.
Photo: Himarerme, Public domain, Wikimedia Commons.

Work of underground water — Karst

- **Karst topography** = landscape of **limestone (soluble rock)** shaped by the **dissolving action of underground water**, with sinkholes, caves and underground drainage. P23 Q7e
- **Process:** rainwater absorbs CO_2 → weak **carbonic acid** → dissolves limestone (calcium carbonate) along joints (**solution**).
- **Erosional features: lapies** (grooved ridges), **sinkholes/dolines** (funnel-shaped hollows), **swallow holes** (where streams vanish underground), **caves/caverns** P23 Q1a(vi): B stalactite is NOT erosional.
- **Depositional features** (dripping water leaves **calcite**): **stalactites** (hang from roof), **stalagmites** (grow up from floor), **pillars/columns** (when they join). P23 Q6a

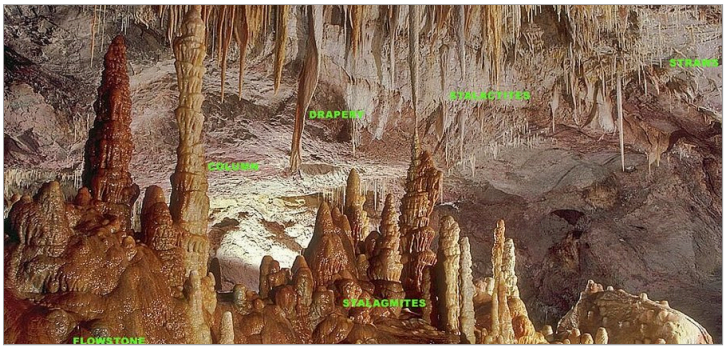


EROSIONAL:
lapies, sinkholes, swallow holes, caves, tunnels

DEPOSITIONAL:
stalactite, stalagmite, pillar (column)

Limestone dissolves in rainwater (carbonic acid) = EROSION; dripping water re-deposits calcite = DEPOSITION

Karst: limestone cave with erosional and depositional features.



A limestone cave: stalactites, stalagmites, columns, flowstone and draperies.
Photo: Dave Bunnell / Under Earth Images, CC BY-SA 2.5, Wikimedia Commons.

Solved questions

P25 Q7(a) [2] How a **stalagmite** forms: water containing dissolved **calcium bicarbonate** drips from the cave roof; when a drop hits the floor it **loses CO_2 and evaporates**, leaving a small deposit of **calcite**. Drop after drop, the deposit builds **upward** into a cone. (If it meets a stalactite growing down, they form a pillar.)

P25 Q7(b) [2] TWO **negative impacts of tourism on karst landscapes**: (1) tourists **touch, break or write on** stalactites and stalagmites (they grow only ~1 cm in 100 years), and **trample** the surface; (2) **waste, litter, oil and sewage** pollute the underground water and cave, and lights/crowds disturb bats and cave life.

P23 Q6(a) [1+1] (i) Figure 1 shows a **pillar (column)** in a limestone cave. (ii) It forms when a **stalactite growing down from the roof and a stalagmite growing up from the floor join**, by repeated deposition of calcite from dripping water.

P23 Q6(b) [2] TWO features formed by wind in arid areas: (1) **Mushroom rock** – wind-blown sand abrades the base of a rock more than the top, giving a narrow stem and wide cap (**erosion**); (2) **Sand dunes** – wind drops sand where it loses speed, forming crescent-shaped barchans or long seif dunes (**deposition**). (Others: yardangs, loess, deflation hollows.)

P23 Q7(e) [1] Karst topography: the unusual landscape that develops on **limestone** when **slightly acidic rainwater and underground water dissolve the rock**, creating sinkholes, caves, underground streams and, where water drips, stalactites and stalagmites.

Quick: P25 d: i **d** · ii **a** · iii **c** · iv **e** · v **b** (f deflation and g erosion unused). P23 vi **B**.

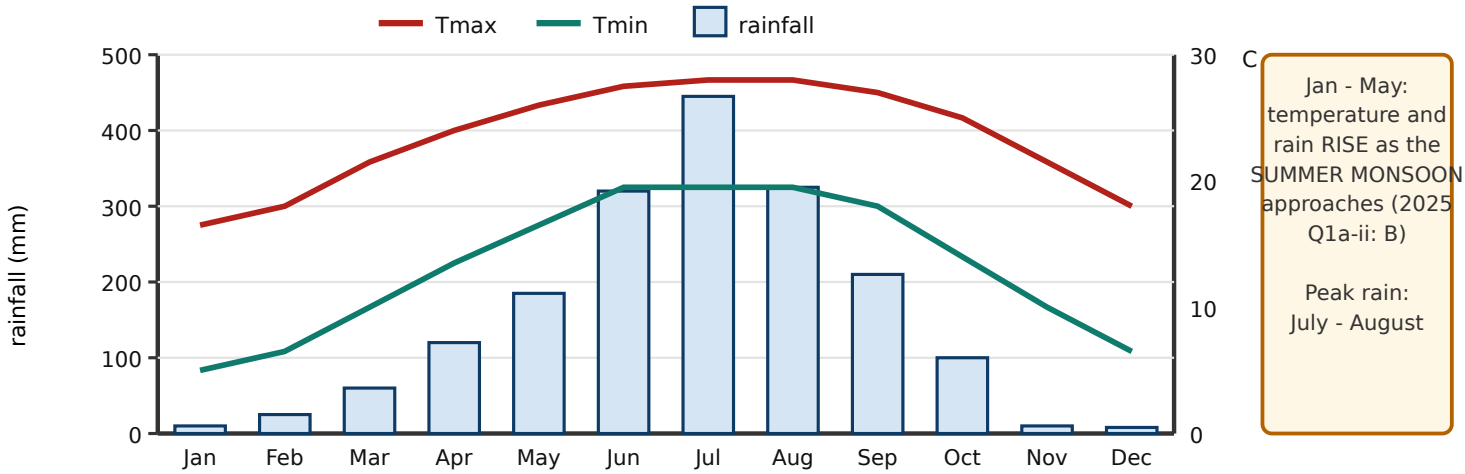
4 · CLIMATE, VEGETATION & HOUSES OF BHUTAN

P25 Q1a(ii,v),c(ii,iii),Q2e,Q7d · P23 Q1a(iii),Q3b,Q5c,d,Q7d

Factors controlling Bhutan's climate

- **Altitude** (the main factor) — Bhutan rises from ~100 m to 7,500 m; temperature falls ~**6.5 °C per 1000 m**. High regions are **cooler** P25 Q1c(ii): lower. Gasa has cool summers and cold dry winters mainly because of its **altitude** P25 Q1a(v): C.
- **Mountains (direction/orientation of ranges):** they block the monsoon winds — **windward southern slopes get heavy rain (5000+ mm), leeward inner valleys and the north are dry** (rain shadow); mountains force air to rise and cool, forming precipitation (orographic). Mountain ranges **affect precipitation** → 2025 b(iv) **TRUE**.
- **Monsoon:** moist south-west monsoon winds from the Bay of Bengal bring most rain in **June–September**. The graph's rainfall rise from January to May shows the **approach of the summer monsoon** P25 Q1a(ii): B.
- **Latitude** (26.7°–28.3° N, sub-tropical) gives high sun angle, but altitude overrides it.
- **Seasons:** winter (Dec–Feb), spring (Mar–May), summer monsoon (Jun–Sep), autumn (Oct–Nov).

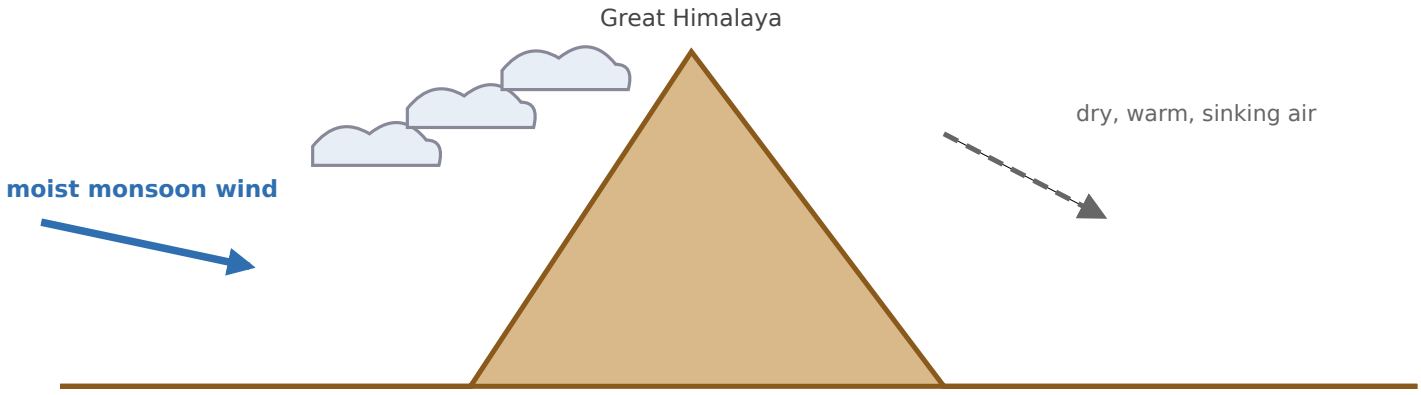
Climate of Bhutan (redrawn, approximate values)



The climate graph from the 2025 paper, redrawn.

climate graph

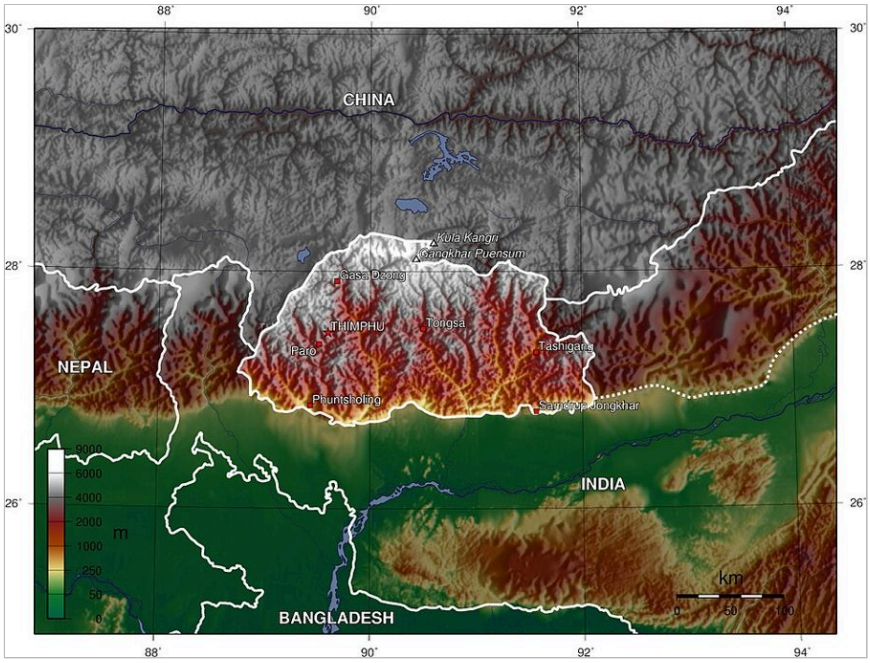
Mountains force air to rise, cool and drop rain on one side (orographic rainfall)



WINDWARD: heavy rain
(southern foothills 5000+ mm)

LEEWARD / RAIN SHADOW: dry
(Paro, Thimphu, north: ~ 500-1000 mm)

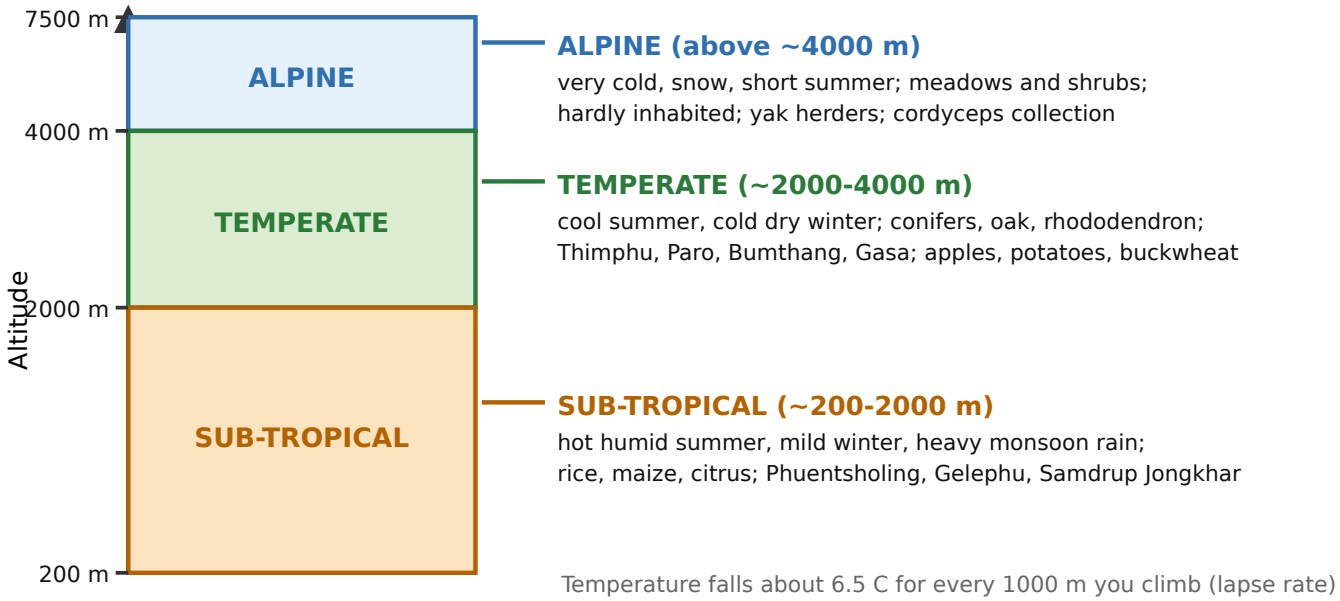
Why the south is wet and the interior dry.



Map: relief of Bhutan — low foothills in the south rising to the high Himalaya in the north.

Photo: Unknown, CC BY-SA 3.0, Wikimedia Commons.

Climatic zones & vegetation



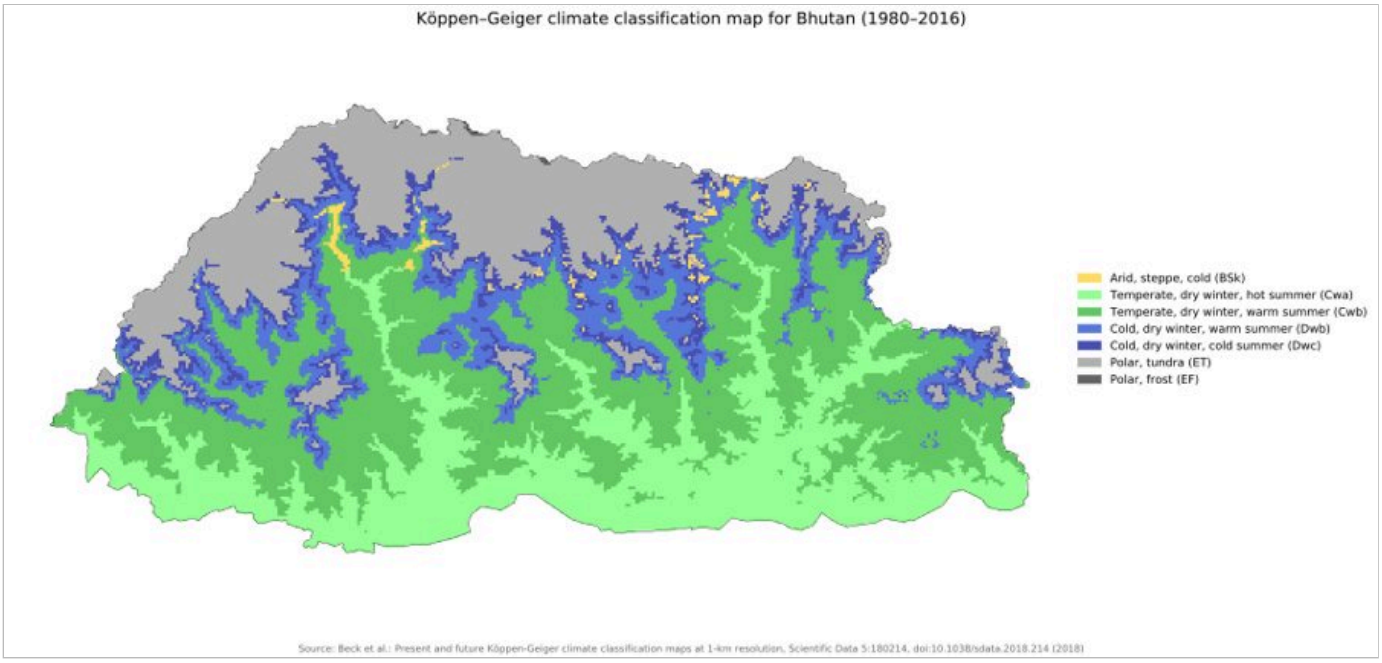
ALPINE (above ~4000 m)
very cold, snow, short summer; meadows and shrubs;
hardly inhabited; yak herders; cordyceps collection

TEMPERATE (~2000-4000 m)
cool summer, cold dry winter; conifers, oak, rhododendron;
Thimphu, Paro, Bumthang, Gasa; apples, potatoes, buckwheat

SUB-TROPICAL (~200-2000 m)
hot humid summer, mild winter, heavy monsoon rain;
rice, maize, citrus; Phuentsholing, Gelephu, Samdrup Jongkhar

Temperature falls about 6.5 C for every 1000 m you climb (lapse rate)

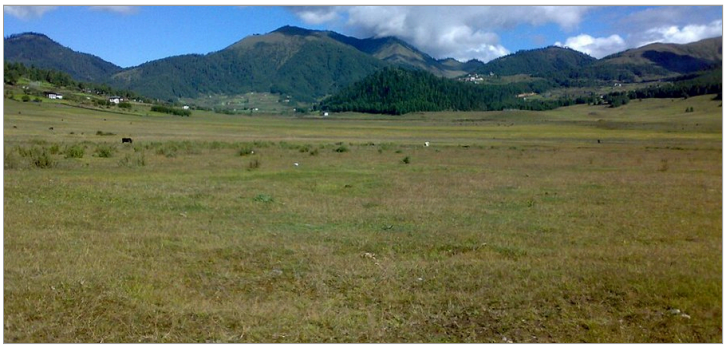
Climatic zones by altitude.



Map: climate zones of Bhutan (Köppen–Geiger).
Photo: Beck, H.E., Zimmermann, N. E., McVicar,..., CC BY 4.0, Wikimedia Commons.



Gangkhar Puensum — the alpine / glacial zone of the high Himalaya.
Photo: Gradythebadger, CC BY-SA 4.0, Wikimedia Commons.

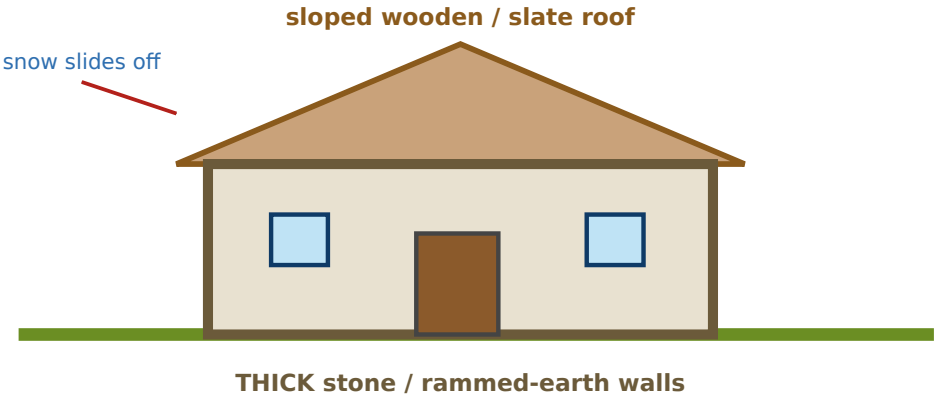


Phobjikha Valley — a broad, cool glacial valley in central Bhutan.
Photo: Phuentsho, CC BY-SA 4.0, Wikimedia Commons.

- **Alpine zone** is hardly inhabited because the climate is **cold throughout the year**, the ground stays **frozen most of the time** and **farming is not possible** P23 Q1a(iii): D. Collecting valuable **cordyceps (Balu-Sulu / yartsa gunbu)** gives highlanders income, so it is **not** a reason for the zone to be uninhabited.
- **Altitude 200–2000 m = sub-tropical zone** P25 Q1c(iii) **(check zone names in your textbook)**.
- **Climate affects plants and animals:** temperature and rainfall decide the vegetation type; animals adapt (thick fur, hibernation, migration) P23 Q3b.

Figure A (P23 Q5c)	Figure B
Alpine meadow / scrub	Temperate coniferous forest
Treeless: grasses, flowers, low shrubs	Dense tall trees (blue pine, fir, hemlock)
Very high altitude; very cold, short growing season	Lower (about 2000–3500 m); cool, more moisture
Thin soil, strong winds	Deeper soil; sheltered slopes
Grazing: yaks; herb collection	Timber, fuelwood, non-timber forest products

Houses & livelihoods



A house for a very cold area (P23 Q7d, P25 Q7d).

Why (Lunana / cold areas):

1 Thick walls store heat and block cold wind
2 Sloped roof sheds snow and rain
3 Light wood/slate roof: local, easy to repair
4 Few, small south-facing windows; low ceilings; animals below, family above

Solved questions

P23 Q5(d) [2] “The Himalayan ranges affect Bhutan’s climate.” **Agree.** (1) They **act as a barrier to the monsoon**: the southern slopes receive heavy rain while the inner valleys are in a rain shadow and dry. (2) Their **high altitude makes the north very cold** and they **shield Bhutan from freezing Siberian winds**; the north–south valleys also create different local climates.

P23 Q5(c) [2] Figure A is alpine vegetation, Figure B is coniferous forest: (1) A = **treeless grassland/shrubs**; B = **dense tall conifers**. (2) A grows at **very high altitude with very cold climate**; B grows at a **lower altitude with a cool, moist climate**.

P25 Q7(d) / P23 Q7(d) [2] Thick-walled houses with sloping roofs (Lunana): (1) **thick stone or rammed-earth walls keep heat inside and block the freezing wind**; (2) **sloped roofs of wood/slate let snow and rain slide off** so the roof does not collapse and does not leak; (also: local, light, easy-to-repair materials). **Design for a very cold place:** thick insulated walls, small south-facing windows, a sloping roof, a tight door, low ceilings and animals kept below the living floor.

P25 Q2(e) [2] Highlander (cordyceps) or southern farmer? Sample answer: **I would live in the south and farm.** (1) The **sub-tropical climate and fertile land** allow farming for most of the year with regular income; (2) the south has **roads, markets, schools and hospitals**, whereas cordyceps income is seasonal, risky and lives in a harsh climate at high altitude. (Either choice is accepted if the two reasons are sound.)

Quick: P25 ii **B**; v **C**; c(ii) **lower**; c(iii) **sub-tropical** (check). P23 iii **D**.

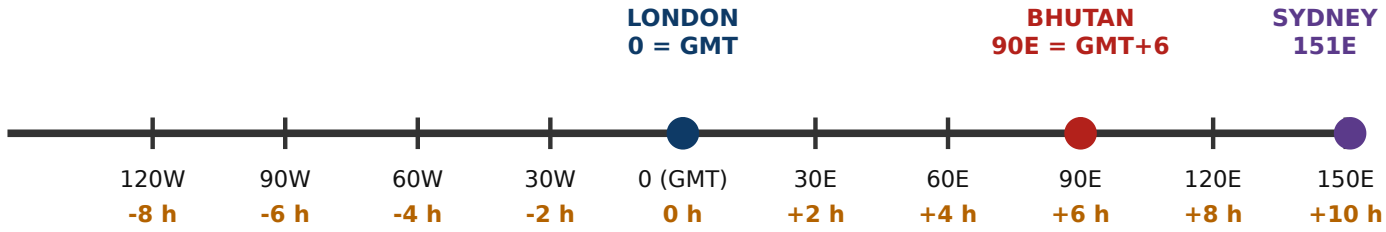
5 · LATITUDE, LONGITUDE & TIME

P25 Q1a(iv),Q3e,Q4a,c · P23 Q1a(ii),c(i),Q3a,Q5b,Q7c

Key ideas

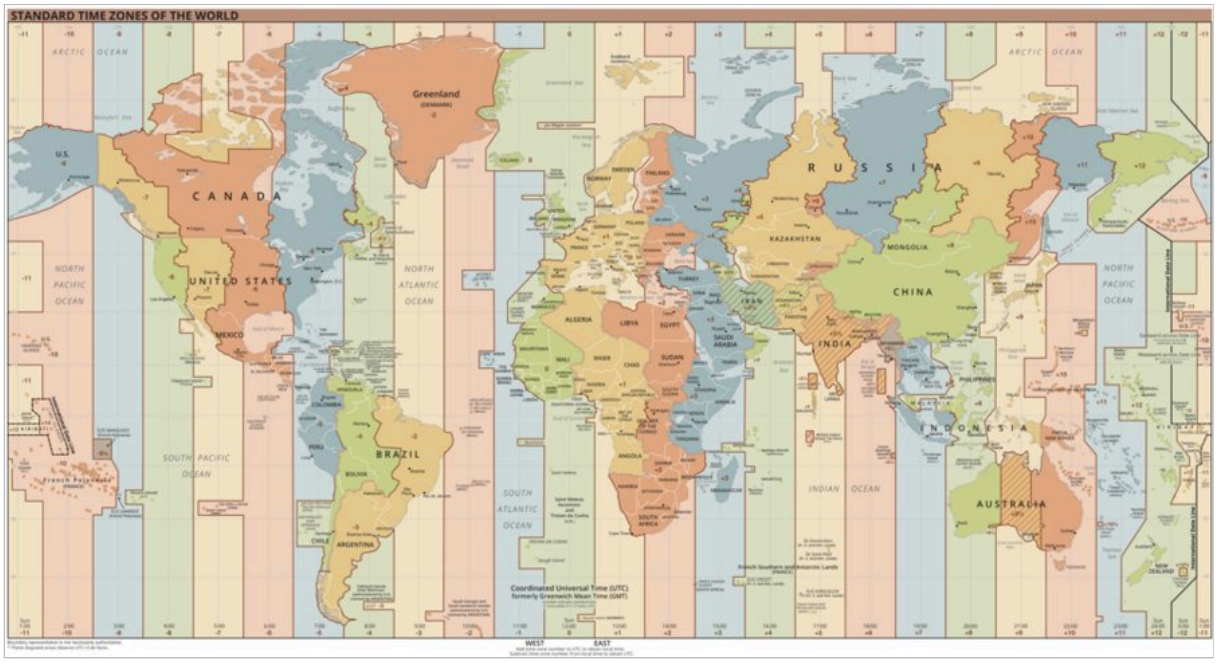
- **Latitude** = angular distance north/south of the Equator (0°–90°); parallels. **Longitude** = angular distance east/west of the **Prime Meridian (0°, Greenwich)**; meridians (0°–180°).
- **Earth rotates 360° in 24 hours → 15° = 1 hour → 1° = 4 minutes.** East is ahead; west is behind.
- **Standard time:** Bhutan **90° E → GMT + 6 h**; India 82½° E → GMT + 5½ h; London (GMT) 0°; Singapore 104° E; Sydney 151° E.
- **Uses of latitude** P25 Q4a: (1) locates a place and its **climate zone** (tropical, temperate, polar); (2) gives **day length and seasons**; (3) used with longitude for **navigation**; (4) measures **north–south distance** (1° ≈ 111 km).
- **GMT** = mean solar time at Greenwich. **Limitations** P25 Q4c: based on **one meridian only** and the Sun, so it varies by place and season; Earth's rotation is not perfectly steady, so it cannot meet the needs of **atomic-precision timekeeping** (GPS, internet, aviation), which now use **UTC**; it ignores **daylight-saving** changes and large countries using several zones.
- **GPS** (Global Positioning System) is an **advanced** method of finding latitude and longitude using satellites → 2023 c(i) **FALSE**.

Earth turns 360 deg in 24 h: 15 deg = 1 hour, 1 deg = 4 minutes



EAST of a place = AHEAD in time (+). WEST = BEHIND (-).
Time difference = (longitude difference x 4) minutes. Local time = GMT + (longitude / 15) h.
Same longitude side: subtract the longitudes; opposite sides (E and W): add them.

Time and longitude.



Map: the world's standard time zones.
Photo: UnaitxuGV, Heitordp and others, Public domain, Wikimedia Commons.

Solving time problems

1. Find the **longitude difference** (subtract if both are E or both W; **add** if one is E and one W).
2. Multiply by **4 minutes** (or divide by 15 for hours).
3. **Add** the time difference for a place to the **east**; **subtract** for a place to the **west**.

Solved questions

P25 Q1a(iv) [1] 9 AM in Bhutan → London: difference 6 h (Bhutan is ahead), so 9 − 6 = **3 AM (C)**.

P23 Q1a(ii) [1] Sydney 151° E, Bhutan 90° E: difference = 61° × 4 = 244 min = 4 h 04 min; Sydney is east (ahead): 5:00 a.m. + 4 h 04 = **9:04 a.m. (B)**.

P25 Q3(e) [2] Singapore time 5:48 PM, Delhi (77° E) 4:00 PM → Singapore is ahead by 1 h 48 min = 108 min. Longitude difference = 108 ÷ 4 = **27°**. Singapore is east: 77° + 27° = **104° E**.

P23 Q3(a) [2] Delhi (70° E) 8 p.m.; Place A 5 a.m. (next day) = 9 hours ahead → Place A is **east**. Difference = 9 × 15 = 135°. Longitude of A = 70° + 135° = 205° E = **155° W**.

P23 Q5(b) [2] X is 20° W, Y is 50° W. **X (20° W) is farther east, so it sees the sunrise first.** Difference = 50° − 20° = 30° × 4 = 120 min = **2 hours** (Y has sunrise 2 hours later).

P23 Q7(c) [2] Changlimithang 6 p.m. (90° E) → London (0°): 90° × 4 = 360 min = 6 h behind → **12 noon (GMT)** on 19 August 2022. (In August London uses British Summer Time, GMT + 1, so broadcasts appear at **1 p.m.**; the standard textbook answer is 12 noon.)

P25 Q4(c) [2] GMT's limits: (1) it is tied to the **Sun at one meridian** and Earth's rotation is irregular, so it is not precise enough for modern needs (satellites, banking, aviation) which use **UTC from atomic clocks**; (2) it **does not suit countries spanning many longitudes** or those that change clocks for daylight saving, causing confusion across time zones.

P25 Q4(a) [2] TWO uses of latitude: (1) it tells which **climatic zone** a place lies in (hot tropics, temperate, cold polar) and the **length of day** and seasons; (2) it is used to **locate places and navigate** (ships, aircraft) and to measure north–south distance.

Quick: P23 c(i) **FALSE** (GPS is advanced). Formula card: **4 min per degree; 15° per hour; Bhutan = GMT+6.**

6 · MAP READING & SURVEY TOOLS

P25 Q1a(i),b(i),e(i–ix),Q2a,Q5c · P23 Q1c(i),e(i–viii)

Topographic maps

- A **topographic map** shows the **natural and man-made features** of an area in detail: relief (contours, spot heights), drainage, vegetation, settlements, roads, trails, boundaries. Survey of Bhutan maps are at **1 : 50,000**.
- **Map numbering (India & adjacent countries series, IAC):** degree sheet (e.g. **78**) → divided into 16 sheets lettered **A–P** (1° × 1°) → each lettered sheet divided into 16 numbered **1–16** (15′ × 15′, the 1:50,000 map). A map is written e.g. **78M/3** (Mongar) or **78M/11** (Trashigang). The numbering of topographic maps is **not the same in every country** → 2025 b(i) **FALSE**.
- **Conventional signs and symbols** (colours and small pictures) P23 Q1e(vii): blue = water, green = vegetation, brown = contours, black = man-made features, red = roads/settlements. They are important because: (1) they give a **universal language** so any user can read any map; (2) they **save space and avoid clutter** of words, so more information is shown clearly; (3) they let us identify features and routes quickly.

Degree sheet 78 (1 deg x 1 deg)

A	**E**	I	M
B	F	J	N
C	G	K	O
D	H	L	P

Sheet 78E split into 16 (15' x 15')

1	5	9	13
2	6	10	14
3	7	**11**	15
4	8	12	16

Map number reads: degree sheet / letter / number = 78E/11

Columns count DOWN: 1-4, 5-8, 9-12, 13-16 so the highlighted cell (3rd column, 3rd row) is 11 (2025 Q1a-i: B)

How a topographic map number is read (2025 Q1a-i).

Scale

Type	Meaning / example
Statement scale	“1 cm represents 0.5 km”
Representative fraction (RF)	1 : 50,000 (no units; same unit on map and ground)
Linear / bar scale	a drawn line showing ground distances

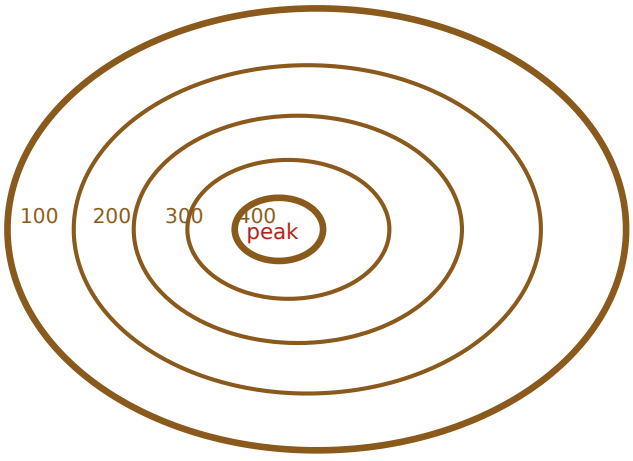
RF → statement scale: divide the denominator by 100,000 to get **km per cm** (denominator in cm ÷ 100,000 cm per km).

RF	Statement scale
1 : 50,000	1 cm = 50,000 cm = 500 m = **0.5 km**
1 : 600,000	1 cm = 600,000 cm = **6 km**

Grid references & contours

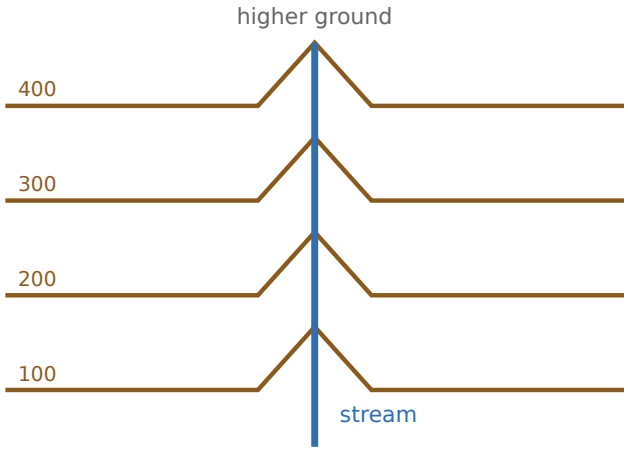
- **Four-figure reference** = a grid square (e.g. **2030**: easting 20, northing 30). **Six-figure** = an exact point: easting (2 digits + tenth) then northing (2 digits + tenth), e.g. **405502**. Rule: **“along the corridor, then up the stairs”** — read the **eastings (left→right) first**, then the **northings (bottom→top)**.
- **Contours** join points of equal height. **Contour interval** = vertical gap between two contours (read from the map’s margin; every fifth is a thicker **index contour**). Close = steep; wide = gentle; V pointing uphill = valley; V pointing downhill = spur; **spot height** = a point with its height printed.
- **Drainage patterns: dendritic** (tree-like), **trellis** (parallel main rivers with right-angled tributaries), **radial** (outward from a peak), **parallel** (steep uniform slope).

Contours of a hill

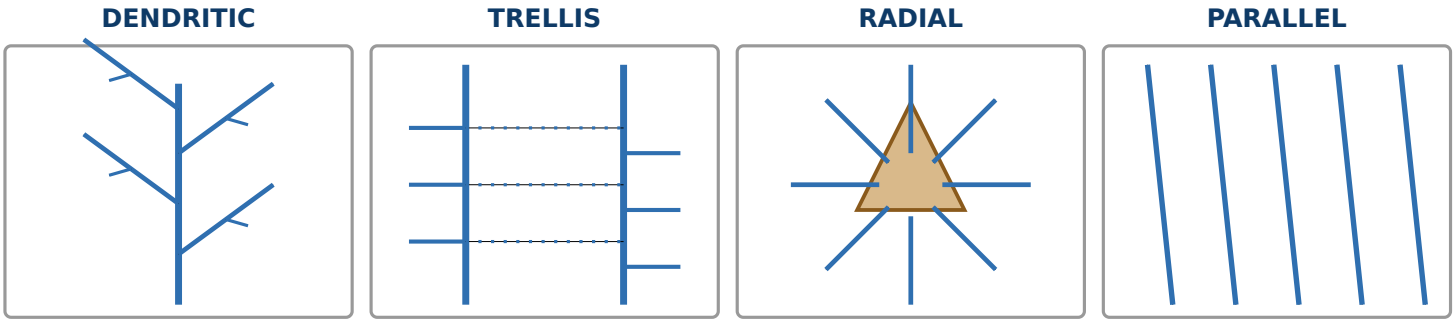


Close together = STEEP slope;
far apart = GENTLE slope

River valley: contours form a V pointing UPHILL



Reading relief from contours.



Common drainage patterns.

Measuring the length of a river

1. Lay a **thread** carefully along every bend of the river on the map.
2. Straighten it and measure with a ruler → map length (cm).
3. Multiply by the scale: at 1:50,000, **length (km) = cm × 0.5**.
   Example: 8.6 cm × 0.5 = **4.3 km**. P23 Q1e(iv) **(map)**

Survey and geospatial tools

- **Plane table survey:** equipment = **plane table (drawing board), tripod stand, alidade, spirit level, plumbing fork and plumb-bob, compass, drawing sheet, pins**.
- **GPS** (satellite positioning), **GIS** and **Google Earth** (satellite images, 3-D terrain, measuring tools).

Solved questions — Sec A

P25 Q1a(i) [1] The figure shows the 1° sheet **78**, sub-sheet **E**, and the highlighted cell is in the 3rd column, 3rd row; the cells are numbered down each column (1–4, 5–8, 9–12...) so it is **11**. Map number = **78E/11 (B)**.

P25 Q1e(i) [1] Index contour interval of 78M/3: read it in the **map margin (legend)**. Survey of Bhutan 1:50,000 maps use a **20 m** contour interval (index contours thick every 100 m). **(map)**

P25 Q1e(ii) [1] Six-figure grid reference for spot heights 1950 and 590: find the dot and number, read the **easting** at the bottom (two digits + estimate of tenths) then the **northing** at the side. e.g. a point 3/10 across square 20 and 6/10 up square 30 is **203306**. **(map — read from your sheet)**

P25 Q1e(iii) [2] Topographic maps help plan **hiking or a campaign**: (1) **contours show the slope and height** so a gentle route, steep climbs and the best camping sites can be chosen; (2) they show **trails, roads, rivers, bridges, settlements and sources of water**, so distances and safe routes can be measured and hazards (cliffs, rivers) avoided.

P25 Q1e(iv) [2] Boosting tourism in the Mongar area: (1) **develop and promote cultural/religious sites (gompas, dzongs, lhakhangs) and nature trails** with signs and trekking routes; (2) **improve roads, homestays and services** so visitors can stay longer; justify: more visitors bring income and jobs to local people. **(map – use the features on your sheet)**

P25 Q1e(v) [2] RF 1 : 600,000 → 1 cm on the map = 600,000 cm on the ground = 6,000 m = **6 km**. **Statement scale: 1 cm represents 6 km.** (The 78M/3 sheet itself, 1:50,000, is 1 cm = 0.5 km.)

P25 Q1e(vi) [2] As DAO, the best area for advanced farming is **flat or gently sloping land close to a river/stream and a road near settlements** (wide contour spacing). Reasons: (1) gentle slope allows **machinery, terraces and irrigation** with less soil erosion; (2) **water and roads/markets nearby** make inputs and sales easy. **(map)**

P25 Q1e(vii) [2] Drainage in 2030 vs 2060 **(map)**: compare patterns — e.g. one square shows **many short streams branching like a tree (dendritic)** on a steep hillside, the other **few, long, widely spaced streams** in a gentle area; or one has a main river and tributaries, the other has none. State the pattern, the stream density and the slope.

P25 Q1e(viii) [2] Large-scale **agro-based industry** in grid square 6060 **(map)**. Judge by: (1) **land**: flat/gentle, wide-spaced contours? (2) **roads/river, settlements and farm land nearby** for labour, raw material and transport. If the square is **steep and forested with few roads → not feasible**; if it has **flat farmland, road and settlement → feasible**.

P25 Q1e(ix) [1] From Kurizampa to Serbum Gompa: mode of transport is the one shown by the **route symbol** between them — a **footpath or mule track → walk**; a motor road → bus/car. **(map)**

P23 Q1e(i)–(viii) The Trashigang 78M/11 sheet is not in the PDF; answer by method: **(i)** altitude of the lhakhang near Dezhuling = the **spot height or contour** beside the symbol; **(ii)** farming at 8040: check **flat land, water, roads, settlement**; **(iii)** civilisations flourish near rivers because of **fertile alluvial soil, water for drinking, transport and trade, flat land and mild climate**; **(iv)** length of Thungthi Ri by thread × 0.5 km; **(v)** compare **dendritic** and **trellis/parallel** patterns; **(vi)** man-made features at 405502 (e.g. **road, building, bridge or lhakhang**) and **one importance** (links villages / worship); **(vii)** signs are important — see above; **(viii)** RF 1:50,000 → **1 cm = 0.5 km**.

Conventional signs & colours (quick table)

Colour / sign	Shows
Blue	rivers, streams, lakes, springs, canals
Green	forest, scrub, grass, orchards, cultivated land
Brown	contours, spot heights, relief features
Black	man-made features: buildings, bridges, boundaries, names
Red	roads, motor roads, built-up areas
Dashed line	footpath / mule track; broken line = boundary

How to answer any map question

Question type	What to write
Suitability (farming, industry, tourism)	1) relief (flat or steep?) 2) water/river 3) roads and settlements 4) vegetation/land use
Compare drainage	name the pattern, say how many streams, slope, where the main river flows
Man-made feature at a grid reference	name it (building, bridge, road) + one use (links, worship, shelter)
Length or distance	thread, ruler, × scale (0.5 km per cm at 1:50,000)
Transport choice	look at the line symbol: road, track or footpath

Types of maps

- **Physical** (relief), **political** (boundaries), **thematic** (rainfall, population), **topographic** (detailed, large scale), **cadastral** (land ownership; Bhutan thram maps).

Solved questions — Sec B

P25 Q2(a) [2] Equipment shown: a **tripod stand**. **Uses:** (1) it **supports the plane table (drawing board)** firmly at a convenient height; (2) its legs can be adjusted to **level the table** and keep it steady during survey.

P25 Q5(c) [1] Google Earth is a powerful geospatial tool because it **shows satellite images and 3-D maps of any place**, helping us **find routes and locations, plan travel, and view changes such as forest loss, glaciers and disasters**. (One reason is enough.)

7 · OUTLINE-MAP WORK: ASIA & BHUTAN

P25 Q1f [4], Q1g [6] · P23 Q1f [4], Q1g [6]

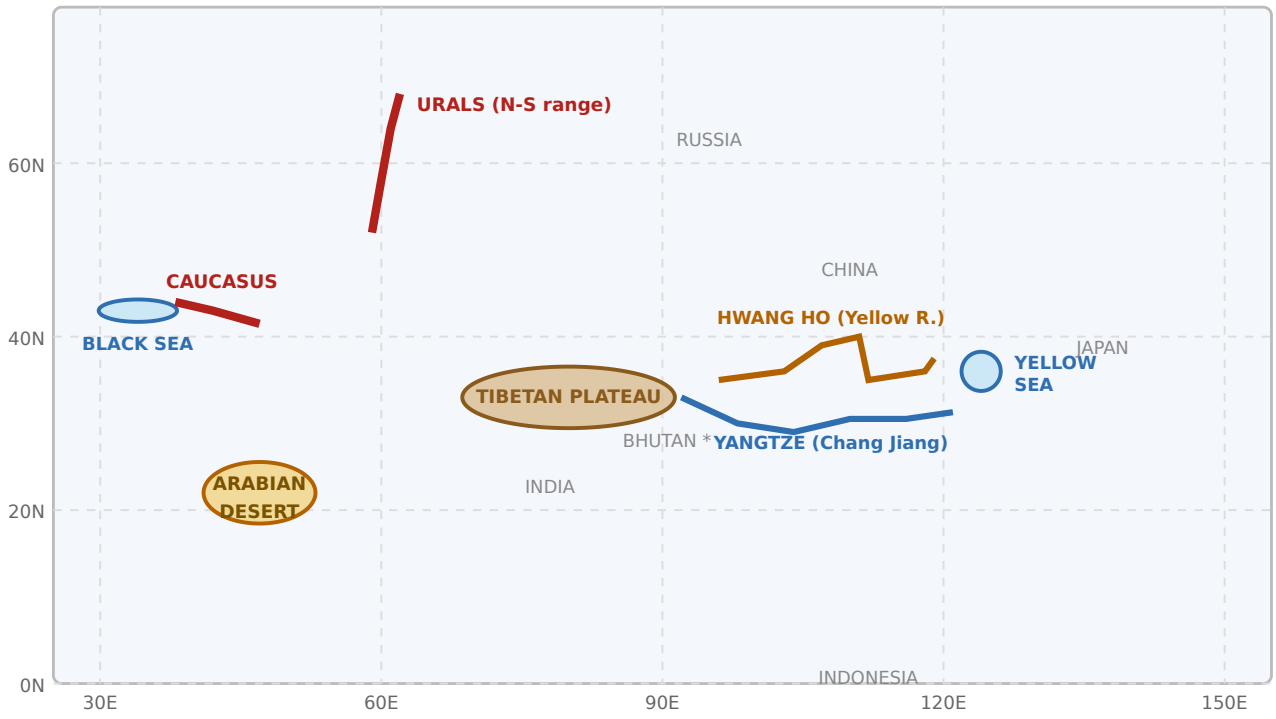
Rules for scoring 10/10

- Use a **sharp pencil**; mark the feature in the **correct place** with the **symbol asked**; write the name beside it (not over lines); shade softly.
- Symbols: **thick line** = river; **∧∧∧** = mountain; **▲** = peak; **● thick dot** = place/dzong; **X** = pass (La) (or][as asked); **DZ** = dzong; **OP** = protected area; **shade** = desert, plateau, sea.

Asia — what to mark and where

Question	Answer and location
P25 (i) largest hot desert in SW Asia	**Arabian Desert** – Saudi Arabia / Arabian peninsula
(ii) mountain Urals ∧∧∧	**Ural Mountains** – Russia, run north–south along ~60° E, boundary of Europe and Asia
(iii) Hwang Ho (thick line)	**Yellow River** – from the Tibetan Plateau through northern China to the Bohai/Yellow Sea
(iv) sea on the border of Europe and Asia	**Black Sea** (between Turkey, Ukraine/Russia and the Caucasus); the Caspian Sea is also on the border **(check which your atlas shows)**
P23 (i) highest plateau	**Tibetan Plateau** (“Roof of the World”, ~4,500 m), north of the Himalayas
(ii) Caucasus ∧∧∧	**Caucasus Mountains** – between the Black Sea and the Caspian Sea (Georgia, Armenia, Azerbaijan, Russia)
(iii) Yangtze river	**Yangtze (Chang Jiang)** – from the Tibetan Plateau east to the East China Sea at Shanghai
(iv) Yellow Sea	**Yellow Sea** – between China and the Korean peninsula

Asia: schematic location guide for the outline-map task (positions approximate)



Where the Asia features are (schematic; use your atlas for exact shapes).



Asia from space — compare with your atlas outline map.
Photo: Unknown, Public domain, Wikimedia Commons.

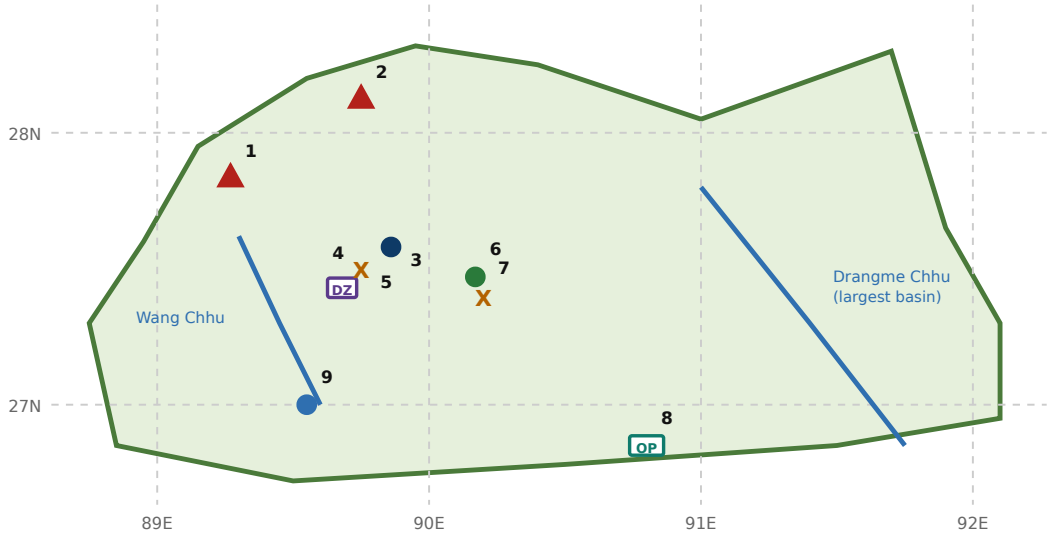
Bhutan — what to mark and where

Question	Answer and location
P25 (i) river with largest drainage basin	**Drangme Chhu** (Manas) in the east **(check)**; the other major rivers: Amo, Wang, Puna Tsang Chhu

(ii) La with 108 Druk Wangyal Chortens	**Dochula** – on the Thimphu–Punakha road (3,100 m)
(iii) Jhomolhari ▲	north-west, on the Thimphu–Paro border (7,326 m)
(iv) first dzong built by Zhabdrung	**Simtokha Dzong** (1629), south of Thimphu — mark “DZ”
(v) black-necked crane festival	**Phobjikha (Gangtey) valley**, Wangdue Phodrang — 11 Nov each year
(vi) Punakha Dzong ●	Punakha, at the meeting of the Pho and Mo Chhu

P23 (i) river of the first hydropower project	**Wang Chhu** (Chhukha Hydropower, Chukha)
(ii) La between Sha and Mangde][	**Pele La** (Wangdue Phodrang – Trongsa) **(check)**
(iii) Masa Gang ∧∧∧	north-west border (Gasa), over 7,000 m
(iv) dzong where the First King was enthroned “DZ”	**Punakha Dzong** (1907)
(v) oldest protected area “OP”	**Royal Manas National Park** (1966), south-central, bordering India
(vi) Tsenda Gang ∧∧∧	north-west border near Masa Gang, over 7,000 m

Bhutan: schematic location guide (NOT to scale; use your atlas for exact positions)



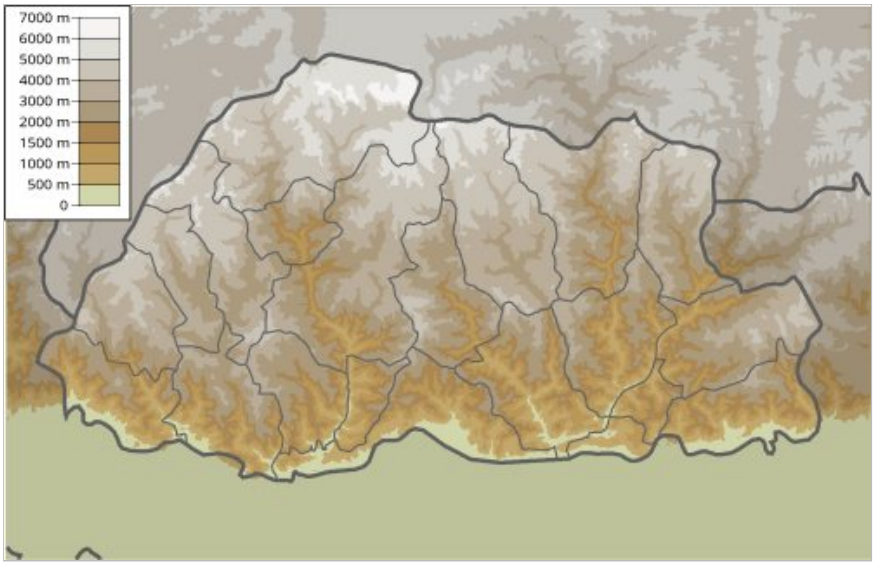
1 Jhomolhari (peak, mark with a triangle)
2 Masa Gang / Tsenda Gang (peaks, NW)
3 Punakha Dzong (First King enthroned; dot)
4 Simtokha Dzong (first dzong, Thimphu; DZ)
5 Dochula Pass (108 chortens; X)
6 Phobjikha / Gangtey (crane festival; dot)
7 Pele La (Wangdue - Trongsa;][ )
8 Royal Manas NP (oldest protected area; OP)
9 Chhukha (first hydropower, on Wang Chhu)

Where the Bhutan features are (schematic).



Map: Bhutan — towns, dzongs, rivers and neighbouring countries.

Photo: Kelisi at English Wikipedia, CC BY-SA 3.0, Wikimedia Commons.



Map: physical features of Bhutan.
Photo: Urutseg, CC0, Wikimedia Commons.

(check) I could not verify every position from an atlas. Positions on the sketches are approximate: always follow the **rivers and borders printed on your outline map**.

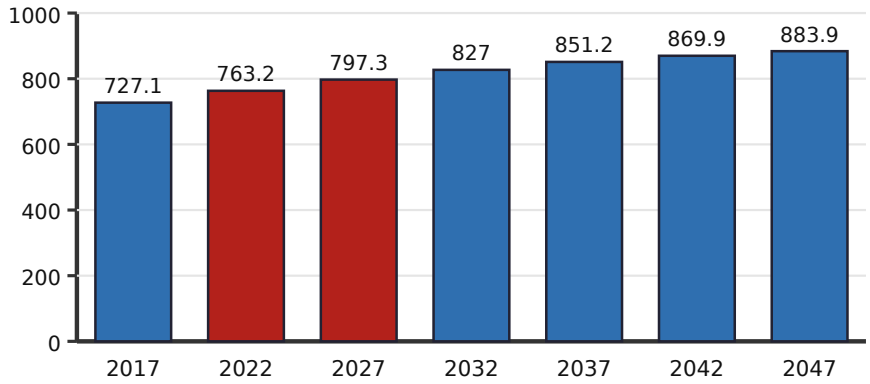
8 · POPULATION & SETTLEMENTS

P25 Q1a(x),Q2b,Q3a,c,Q4b,e,Q6c · P23 Q1a(viii,ix),c(iii,v),Q3d,Q4a,b,Q7a

Key ideas

- **Population change** = births − deaths + (immigrants − emigrants). **Growth rate** = % change per year. Bhutan's rate fell from **1.09% (2012) to 0.64% (2022)**; the world's rate from 1.19% (2015) to 0.91% (2024). Projected Bhutan population: 727,100 (2017) → 797,300 (2027) → 883,900 (2047).
- **Population projection** lets planners **estimate how demographics will develop** (schools, jobs, housing) → 2023 c(iii) **TRUE**.
- **Life expectancy** (70.2 years in 2017) rose through **better healthcare, education and living standards/clean water** P25 Q1a(x): D = i, ii, iv. As it rises, the **mortality (death) rate decreases** P23 Q1a(ix): B.
- **Distribution:** uneven — dense in the west (Thimphu, Paro, Chukha) and southern foothills; sparse in the high north and east. Causes: relief, climate, roads, jobs and services.
- **Migration: internal** (within the country, e.g. village to village in the same dzongkhag, rural → urban) P23 Q1c(v): TRUE; **international** (to other countries). **Push** factors (poverty, no jobs, wildlife damage) and **pull** factors (jobs, services). **Gungtong** = households left empty when families move out of villages.
- **Central Place Theory** (Walter Christaller, 1933): a town is a **central place** serving its surrounding area; **low-order goods** (salt, rice) are bought at the nearest small place so people **travel short distances** for them, **high-order goods** (hospital, university, car) are bought at larger towns, so people travel farther. Hexagonal market areas. **Threshold** (minimum demand) and **range** (maximum distance people travel).
- **Incorrect statement** about CPT: "the number of larger towns and smaller towns is roughly equal" — in fact **smaller towns are many, larger towns few** P23 Q1a(viii): D. Correct: same-size towns evenly spaced; removing a town readjusts the system; larger towns are spaced farther apart.

Projected population of Bhutan (thousands)

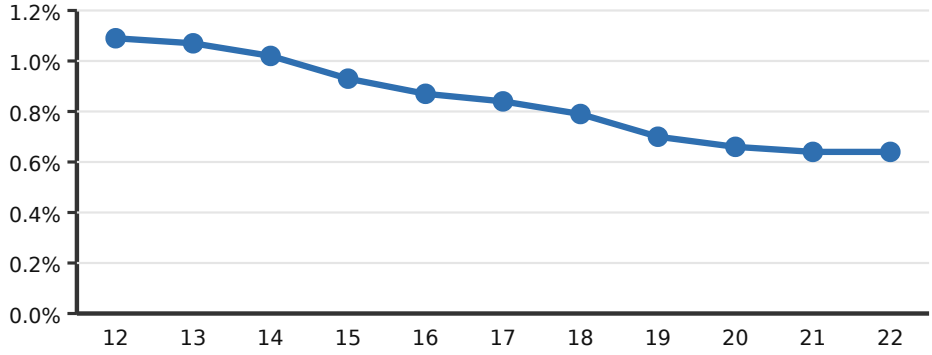


2025 Q3(a): growth 2022 -> 2027
797.3 - 763.2 = 34.1 thousand
= 34,100 persons
% growth = 34.1 / 763.2 x 100
= 4.47 % (about 4.5 %)
(about 0.9 % a year)

Projected population (2025 Q3a) and the calculation.

bar graph

Population growth rate of Bhutan 2012-2022 (%)

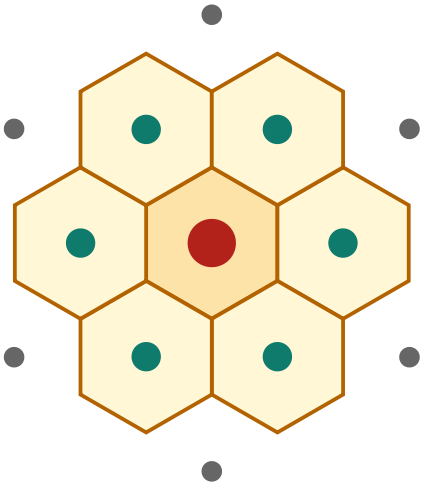


Falls from 1.09% to 0.64%
If it continues:
- ageing population
- fewer workers, labour shortage; more imports
- higher dependency
- fewer students: school closures; slower growth

Growth rate 2012–2022 and consequences (2025 Q6c).

line graph

Christaller's Central Place Theory



- City / large town (high-order goods: hospital, university)
- Town (medium goods)
- Village (low-order: salt, rice, vegetables)

Key ideas:
- people travel SHORT distances for low-order goods and FAR for high-order
- hexagonal market areas (no gaps/overlap)
- many villages, fewer towns, few cities
- THRESHOLD = minimum customers
- RANGE = maximum distance travelled

Central Place Theory.

Solved questions

P25 Q3(a)(calc) [2] Growth 2022 → 2027 = 797.3 − 763.2 = **34.1 thousand (34,100 people)**; as a percentage: 34.1 ÷ 763.2 × 100 = **4.47 %** in five years.

P25 Q2(b) [2] TWO factors for the fall in world population growth: (1) **lower birth rates** from **family planning/contraception and women's education and employment**; (2) **urbanisation and higher cost of living**, so couples marry later and have fewer children (also fewer child deaths, so parents need fewer births).

P25 Q3(c) [2] Central Place Theory: people prefer to **spend less time and money on travel**, so for **cheap daily necessities such as salt and rice** (low-order goods, bought often) they go to the **nearest small village shop/market**; **specialised goods and services** (hospital, college) are found only in larger towns, which are fewer and farther apart. So small centres are many and close, big ones few and far.

P25 Q4(b) [2] As Prime Minister, to settle people in sparsely populated areas: (1) **decentralise services and jobs** — build schools, hospitals, roads, electricity and internet, and set up **agro- and cottage industries** in these areas; (2) give **incentives**: land, housing, tax benefits, higher wages and loans for people who settle there.

P25 Q4(e) [2] 42,829 Bhutanese abroad: **wise, if planned.** (1) They get **better jobs and income and send remittances**, which support families and the economy; (2) they gain **skills and experience** that can be brought back. (Against: **brain drain** and labour shortage — so the policy should be to return with skills.)

P25 Q6(c) [2] If the growth rate keeps falling: (1) an **ageing population** — more old people and a higher dependency ratio, more pressure on health and pensions; (2) a **shrinking workforce and labour shortage**, fewer students (school closures) and slower economic growth, so Bhutan may need immigrant workers.

P23 Q3(d) [2] To reduce **Gungtong**, besides services and wages: (1) **create employment in the village** — agro-processing, cottage industries, ecotourism and homestays, farm cooperatives; (2) **commercialise farming**: market access, machinery, irrigation, protection from wildlife and crop insurance; also awareness and rural credit/youth farm schemes.

P23 Q4(a) [2] International immigrants to fill the labour shortage: **Agree, in a controlled way.** (1) They **supply workers** in construction, farming and services and keep the economy growing; (2) they bring **skills and investment**. (Concern: culture, security and population limits → regulated by permits.) *(Disagree is also accepted with two reasons.)*

P23 Q4(b) [2] TWO criticisms of CPT: (1) it assumes an **unrealistic flat, uniform plain** with the same resources and population everywhere — **mountains, rivers and uneven wealth** upset the pattern; (2) it **ignores transport and modern shopping** (roads, vehicles, online shopping), and people do not always choose the nearest centre.

P23 Q7(a) [2] If Thimphu's population keeps growing (150,595 by 2027): (1) **housing shortage, high rents and squatter settlement**; (2) **traffic congestion, pollution and pressure on water, electricity and waste services**; also unemployment and crime.

Quick: P23 viii **D**; ix **B**; c(iii) **TRUE**; c(v) **TRUE**. P25 x **D**.

9 · AGRICULTURE, LAND & FOOD SECURITY

P25 Q2e,Q3b,Q4d,Q5e · P23 Q1a(vii),d(v),Q3c,Q6d

Key ideas

- **Subsistence farming** = small-scale farming for the family's own food using **primitive tools** and human/animal labour (the past type in Bhutan) P23 Q1d(v): subsistence. **Commercial farming** = large-scale, machinery, hybrid seeds, for sale.
- **Land fragmentation** = farm land divided into small scattered plots as it is shared among children. **Best method: farm cooperatives** (pool land and work together) P23 Q1a(vii): A; also land consolidation.
- **Food self-sufficiency:** Bhutan still imports rice, edible oil and vegetables. **Challenges:** shortage of farm labour (rural–urban migration), **human–wildlife conflict**, small fragmented steep fields, limited irrigation, **climate change**, **pests**. **Ways:** farm mechanisation, electric fencing, irrigation, cooperatives, markets, climate-smart methods, youth in agriculture.
- **Pests:** fall armyworm (maize, since 2019) and Giant African Land Snail (vegetables, since 2010). **Strategies: Integrated Pest Management**: early monitoring and reporting, **pheromone/light traps**, hand-picking and destroying, **biological control** (natural enemies, neem), crop rotation/early planting, quarantine and farmer training.
- **Climate-smart agriculture (CSA): seed saving, crop rotation, terracing**, water harvesting, agroforestry. It raises productivity by keeping soil fertile, saving water and soil and reducing risk from weather.

Solved questions

P25 Q3(b) [2] Punakha farmers face challenges adopting modern technology: (1) fields are **small, terraced and fragmented**, so **machines such as power tillers and harvesters cannot be used easily**; (2) **shortage of labour, money, training and extension support** (and crop damage by wild animals) make farmers hesitate to take risks with new methods.

P25 Q4(d) [2] Strategies against fall armyworm and snails: (1) **monitor fields early and control biologically/with traps** (pheromone traps, neem sprays, natural enemies), hand-pick and destroy snails; (2) **awareness and training** for farmers in integrated pest management, crop rotation and quarantine so pests do not spread.

P25 Q5(e) [2] Does climate-smart agriculture increase productivity? **Yes.** (1) **Crop rotation and seed saving** keep soil fertile, cut pests and give locally adapted seed at low cost; (2) **terracing** stops soil erosion and holds water on steep slopes so yields stay stable under changing climate.

P23 Q3(c) [2] The picture shows **manual paddy transplanting** (traditional farming). **Drawbacks:** (1) it is **labour-intensive, slow and tiring** (bending for hours) and needs many workers; (2) yields and income are **low** because of small fields and primitive tools, so young people leave farming.

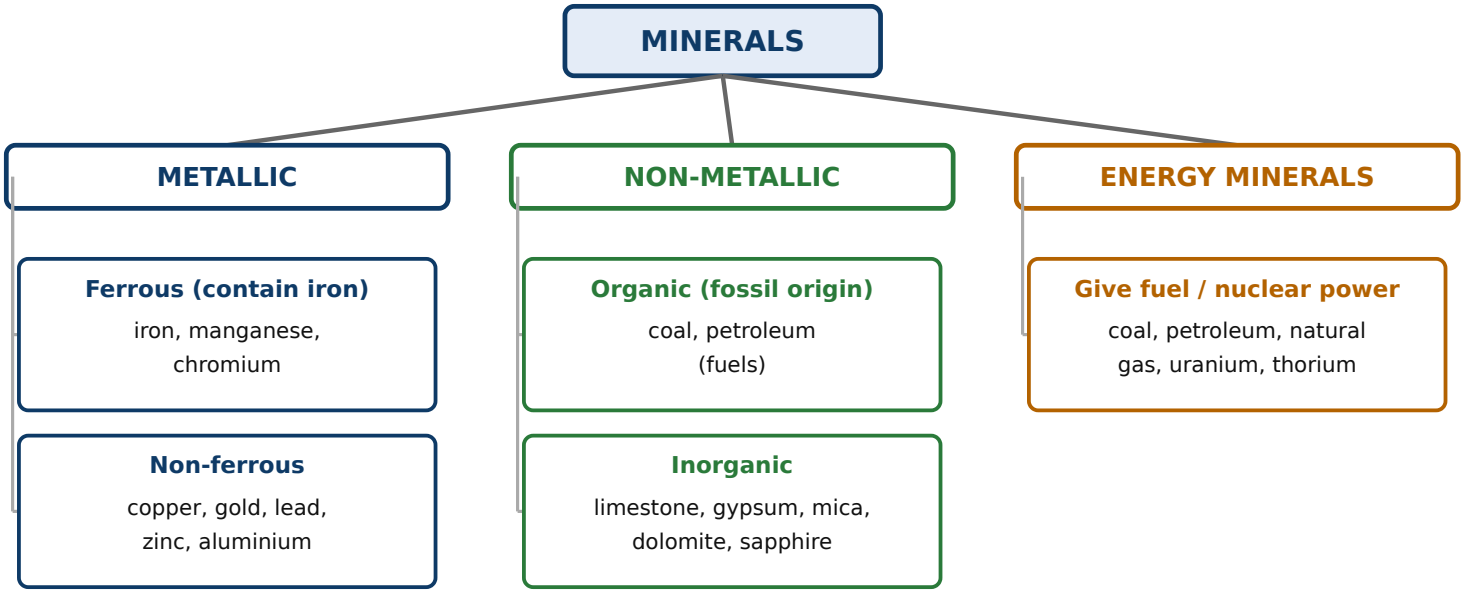
P23 Q6(d) [3] Bhutan is not self-sufficient in rice, oil and vegetables. TWO challenges and solutions: (1) **shortage of labour** as people migrate → introduce **machinery**, labour-saving tools and cooperatives; (2) **crop damage by wild animals/pests and small fragmented holdings → electric fences, IPM**, land pooling and irrigation.

Quick: P23 vii **A**; d(v) **subsistence farming**. 2025 Q2(e) is answered in Ch 4.

10 · MINERALS, INDUSTRY & ENERGY

P25 Q1a(viii),b(ii),Q2d,Q5d,Q6a,Q7c · P23 Q1b,d(i),Q2a,b,Q4c

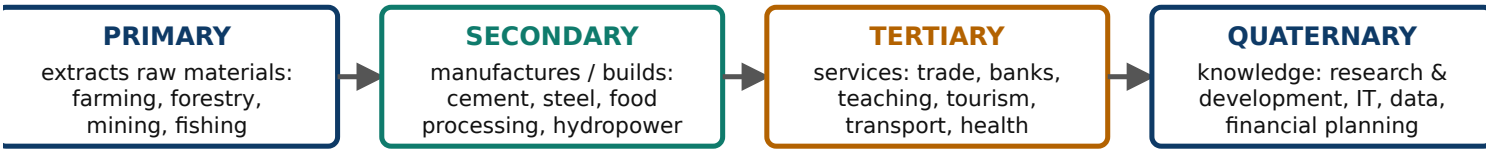
Minerals



Classification of minerals.

- **Matching** P23 Q1b**:** ii **Ferrous metallic → a) Manganese**; iv **Non-ferrous metallic → b) Copper**; iii **Energy minerals → e) Thorium** (radioactive fuel); i **Organic non-metallic → d) Petroleum** (fossil origin); v **Inorganic non-metallic → c) Limestone (check: limestone/sapphire both inorganic; f) Sapphire is a gemstone)**.
- **Mining in Bhutan:** dolomite, limestone, gypsum, coal, quartzite, talc; used in cement, steel, ferro-silicon and calcium carbide. Minerals are **non-renewable**.

Industry



Four sectors of industry.

- **Quaternary industries** = knowledge-based work: **research and development, information technology** P25 Q1a(viii): D; a financial planner / digital stockbroker is in the **quaternary** sector P23 Q1d(i) (high salary).
- **Factors for locating an industry** P25 Q6a: **raw materials, power/water, labour, market, transport and roads, capital, government policy, environmental impact**.
- **Smart factory:** uses sensors, the Internet of Things, robots, data and AI to run production automatically.
- **Solar energy** in Bhutan is a useful renewable source, but **hydropower, not solar, is the most reliable energy source** (solar depends on sunshine, which monsoon clouds and mountain shadows reduce) → 2025 b(ii) **FALSE (check)**.

Solved questions

P23 Q2(a) [2] FOUR negative impacts of mining: (1) **deforestation and habitat loss**; (2) **air pollution** (dust, blasting fumes) and **noise**; (3) **water pollution and siltation** of rivers by waste; (4) **soil erosion, landslides and scarred land**; also health problems of workers and nearby people.

P23 Q2(b) [2] FOUR positive influences of technology on industries: (1) **higher production and productivity** (machines work faster); (2) **better quality and lower cost**; (3) **safer work** and less labour needed in dangerous tasks; (4) **new products, new jobs and wider markets** (online sales); also less waste and energy saved.

P23 Q4(c) [2] Is it profitable for Bhutan to set up many mining industries? **No, not many.** (1) Deposits are **small and scattered in steep terrain**, so mining and transport costs are high; (2) the **damage to forests, rivers and tourism-based biodiversity** costs more than the profit and goes against the pledge to protect nature (60% forest cover).

P25 Q2(d) [2] "Mining gives huge benefits at the cost of the environment." **Agree.** (1) Mining gives **revenue, jobs and raw material for industries** (cement, steel); (2) but it **destroys forests, pollutes air and water, causes landslides and erosion** and the minerals are non-renewable.

P25 Q5(d) [2] How a smart factory reduces waste and improves resource use: (1) **sensors and AI monitor materials, energy and machines in real time**, so exact amounts are used and over-production is avoided; (2) **predictive maintenance and automatic quality checks** reduce breakdowns and defective products; waste is recycled back into production.

P25 Q6(a) [2] TWO most important factors before setting up an industry: (1) **availability of raw materials, power and water**; (2) **market and transport** (nearness to customers, roads and railways). Also labour, capital and government policy.

P25 Q7(c) [2] Long-term impacts of forest-based industries on forest ecosystems: (1) **over-harvesting causes deforestation**, loss of habitat and biodiversity and weak regeneration; (2) **soil erosion, landslides, loss of water sources and carbon storage**, which disturb the whole ecosystem (unless felling is sustainable and replanted).

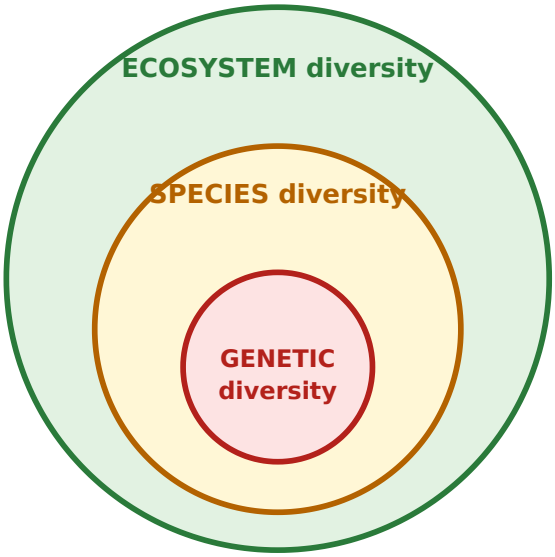
Quick: P25 viii **D**; b(ii) **FALSE** (check). P23 b: i **d** · ii **a** · iii **e** · iv **b** · v **c** (check); d(i) **quaternary**.

11 · BIODIVERSITY, ENVIRONMENT & TOURISM

P25 Q1a(vii),b(v),Q6d,Q7e · P23 Q1c(ii),Q3e,Q6c,e

Key ideas

- **Biodiversity** = variety of life on Earth at three levels: **genetic, species and ecosystem diversity**. Bhutan's **wide variety of habitats** (subtropical forests to alpine meadows) is **ecosystem diversity** P25 Q1a(vii): C; ecosystem diversity results from **species interactions and habitat variation** → 2023 c(ii) **TRUE**.
- **Biodiversity hotspot** = a region with a **very large number of unique (endemic) species that is threatened** by habitat loss → 2025 b(v) **TRUE**. Bhutan lies in the **Eastern Himalaya hotspot**; about 70% forest and over 50% of the land in protected areas and corridors.



Genetic diversity
variety of genes within ONE species
e.g. many local rice, maize and potato varieties; different yak breeds

Species diversity
number/variety of species in an area
e.g. tiger, takin, red panda, black-necked crane, 5,500+ plant species

Ecosystem diversity
variety of habitats and ecosystems
e.g. subtropical forest, temperate forest, alpine meadow, wetlands

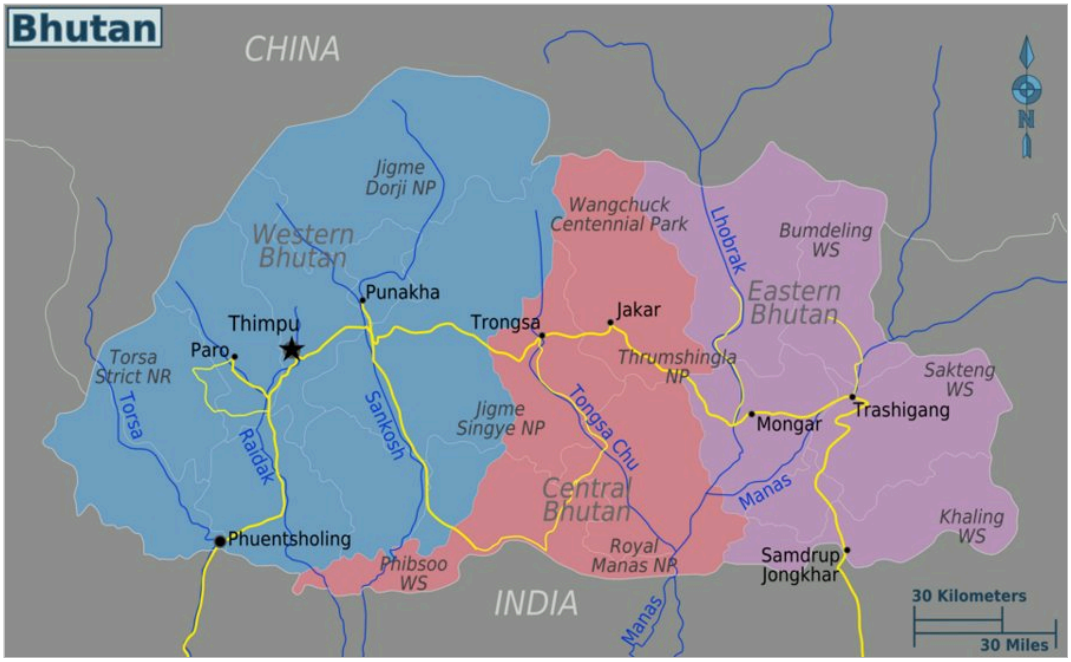
Three levels of biodiversity.



Takin — Bhutan's national animal.
Photo: Gerd Eichmann, CC BY-SA 4.0, Wikimedia Commons.



Black-necked crane — a winter visitor to Bhutan's high valleys such as Phobjikha.
Photo: Sumeetmoghe, CC BY-SA 4.0, Wikimedia Commons.



Map: Bhutan's regions and protected areas.
Photo: Peter Fitzgerald, CC BY 3.0, Wikimedia Commons.

Species diversity (P25 Q6d)	Genetic diversity
Variety of **different species** in an area (number and abundance)	Variety of **genes within one species**
e.g. tiger, takin, red panda, cranes, orchids in one forest	e.g. many varieties of red rice, potato or buckwheat
Higher level; shows richness of an area	Helps a species adapt to disease and climate change

- **Biodiversity and climate change** P23 Q3e: forests are **carbon sinks** that absorb CO_2; healthy forests, wetlands and soils **regulate water, reduce floods and landslides**, and **genetic diversity lets species adapt** to changing climate.
- **Threats:** deforestation, mining, roads/hydropower, vehicles and pollution, poaching, invasive species, climate change.
- **Sustainable Development Fee (SDF)** for tourists (US$ 65 → 200 proposed) supports **"high value, low volume" tourism**.

Solved questions

P25 Q7(e) [2] A **biodiversity protection plan** for my community: (1) **protect and plant forests** — community forests, no hunting or poaching, fire protection and restricted grazing near habitats; (2) **awareness and sustainable use** — school clubs and village meetings, proper waste disposal, ecotourism rules, harvesting of herbs/NTFPs only in quantities that regrow.

P23 Q3(e) [2] Biodiversity and climate change: (1) **forests and plants absorb carbon dioxide and store carbon**, slowing global warming; (2) **diverse ecosystems are more resilient** — they protect against floods, droughts and landslides and provide food and water; genetic diversity helps species adapt.

P23 Q6(c) [2] SDF raised from US$65 to US$200: (1) **fewer tourists** visit, so there is **less pressure, waste and disturbance** in forests, trails and wildlife areas; (2) **the extra money funds conservation** — national parks, community projects, protection of species and carbon-neutral development.

P23 Q6(e) [1] As Director of the National Environment Commission, TWO measures on the rising number of vehicles: (1) **promote electric vehicles** through tax and import incentives and charging stations; (2) **improve public transport and limit private vehicles** (quotas, emission tests, car-pooling, walking/cycling lanes).

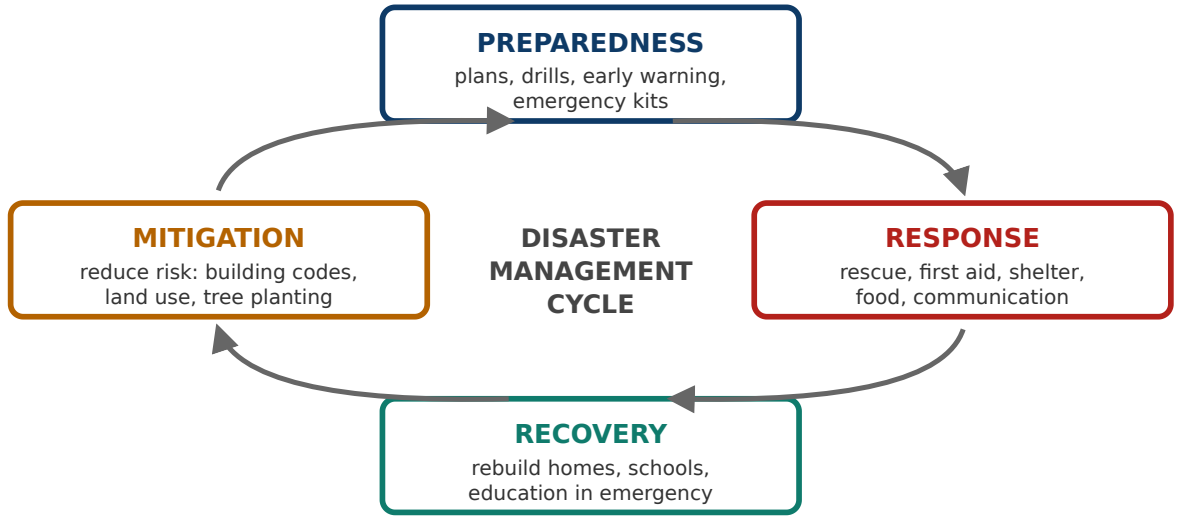
Quick: P25 vii **C**; b(v) **TRUE**. P23 c(ii) **TRUE**.

12 · NATURAL DISASTERS & SAFETY

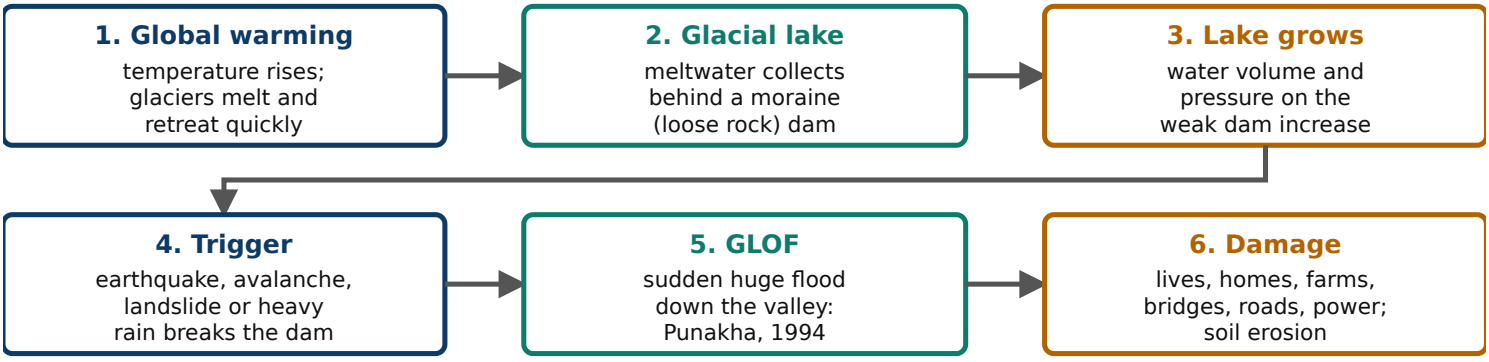
P25 Q1a(ix),c(v),Q3d,Q6b,e · P23 Q1a(x),d(ii),Q2c,Q4d,Q7b

Key ideas

- **Hazard** = a threat; **disaster** = a hazard causing great loss of life and property. **Natural**: earthquake, flood, landslide, GLOF, storm, forest fire. **Disaster management cycle: preparedness, response, recovery, mitigation**.
- **Earthquakes:** Bhutan lies in an **active seismic zone** because the Indian plate presses against Eurasia (e.g. 2009 Mongar earthquake). **Before:** secure heavy furniture, prepare a kit, practise **Drop–Cover–Hold On**; **during:** drop, take cover under a strong table and hold on, stay away from glass and walls, in the open go to open ground; **after:** expect aftershocks.
- **Glacial Lake Outburst Flood (GLOF):** a lake dammed by moraine breaks suddenly. The **Lugge Tsho GLOF of 1994** caused heavy damage in **Punakha** P25 Q1a(ix): D. The disaster caused by **retreating glaciers** due to global warming is a GLOF P25 Q1c(v). **Seismic activity is a natural trigger, not a human-made cause** (human-made: global warming, deforestation, industries) P23 Q1a(x): B.
- **Flash floods and landslides in the southern foothills: very heavy monsoon rain (5000+ mm)**, **steep slopes with young, fragile Siwalik rocks** and loose soil, and **deforestation / road construction**; short steep rivers rise very quickly.
- **Forest fires:** >1,400 incidents in ten years; mostly human-caused in dry winter/spring.
- **Electric fires: unplugging appliances when not in use** and safe wiring reduce **fire** disasters P23 Q1d(ii): fire.
- **Education in Emergency (EiE):** restarting teaching, play and counselling soon after a disaster.



Disaster management cycle.



How a GLOF happens.

Solved questions

P25 Q3(d) [2] TWO methods of fighting **forest fires**: (1) **prevention and early detection** — fire lines/breaks, watch towers, patrols, drones/satellite alerts, awareness and bans on open burning; (2) **quick suppression** — trained community fire-fighting teams with beaters and water, backed by the forest department and the use of controlled burning of fuel before dry season.

P25 Q6(b) [2] Earthquake-safe buildings: (1) **follow the building code**: strong foundation, **reinforced concrete bands/ring beams** at floor and roof level, and **simple symmetrical shapes**; (2) **use light, flexible materials** (timber frames, light roofs), avoid heavy roofs and soft storeys, and **retrofit old buildings**; traditional *dhajji / ekra* walls are flexible.

P25 Q6(e) [2] Technology and life-saving skills in disasters: (1) **early-warning systems, phone alerts, social media and apps** warn and guide people quickly; (2) **GPS, drones, thermal cameras and satellite images** locate victims and assess damage for rescue; also **first-aid videos and online training**.

P23 Q2(c) [2] ONE strategy for earthquakes: **Drop, Cover and Hold On** during the shaking — drop to the floor, cover your head and neck under a sturdy table and hold on until it stops; **before:** keep an emergency kit and know safe spots and exits and fix heavy furniture to walls.

P23 Q4(d) [2] Flash floods and landslides in the southern foothills: (1) **very heavy monsoon rainfall** quickly fills the short, steep rivers; (2) **steep slopes with loose, weak Siwalik rock and soil**, made worse by **deforestation and road cutting**, slide down easily.

P23 Q7(b) [3] “Education in emergency should be carried out by every school.” **Agree.** (1) It **keeps learning going**, so children do not lose a year of education; (2) schooling gives **safety, routine and psychosocial support**, helping children cope with fear and loss; (3) **children are protected** from abuse, child labour and early marriage, and schools can teach **disaster awareness** and act as shelters/centres for the community.

Quick: P25 ix **D** (Lugge Tsho GLOF 1994); c(v) **GLOF**. P23 x **B**; d(ii) **fire**.

SECTION A ANSWER KEY — 2025 & 2023

every MCQ, true/false, fill-in and match in one place · (check) = reasoned answer; (map) = needs the map sheet

2025 Paper — Q1(a) MCQ [10]

No.	Ans	Why (short)
i	**B**	78E/11 (3rd column, 3rd row)
ii	**B**	summer monsoon is approaching (Jan–May rise)
iii	**A**	Georges Lemaître proposed the Big Bang
iv	**C**	9 AM Bhutan − 6 h = 3 AM London
v	**C**	Gasa's climate is set by altitude
vi	**B**	collision of tectonic plates
vii	**C**	wide variety of habitats = ecosystem diversity
viii	**D**	R&D and IT = quaternary
ix	**D**	Lugge Tsho GLOF, Punakha 1994
x	**D**	healthcare, education, living standards (i, ii, iv)

2025 — (b) true/false, (c) fill-in, (d) match

Part	Answers
(b) i–v	i **FALSE** · ii **FALSE** (check: hydropower is the most reliable) · iii **FALSE** (that is the solar nebula theory) · iv **TRUE** · v **TRUE**
(c) i–v	i **accretion** · ii **lower** · iii **sub-tropical** (check) · iv **Pangaea** · v **glacial lake outburst flood (GLOF)**
(d) i–v	i **d** surface creep · ii **a** deposition · iii **c** saltation · iv **e** abrasion · v **b** suspension
(e) map	(i) contour interval from the legend (20 m typical) (map) · (v) **1 cm = 6 km**

2023 Paper — Q1(a) MCQ [10]

No.	Ans	Why (short)
i	**B**	a wider gap between galaxies
ii	**B**	9:04 a.m. (61° × 4 = 4 h 04 ahead)
iii	**D**	collecting Balu-Sulu is not a reason against living there
iv	**C**	converging Indian and Eurasian plates
v	**D**	asthenosphere
vi	**B**	stalactite is depositional, not erosional
vii	**A**	farm cooperatives
viii	**D**	larger towns are fewer, not equal in number
ix	**B**	mortality decreases
x	**B**	seismic activity is natural, not human-made

2023 — (b) match, (c) true/false, (d) fill-in

Part	Answers
(b) i–v	i **d** petroleum (check) · ii **a** manganese · iii **e** thorium · iv **b** copper · v **c** limestone (check)
(c) i–v	i **FALSE** (GPS is advanced) · ii **TRUE** · iii **TRUE** · iv **FALSE** (more greenhouse gas traps MORE heat) · v **TRUE**
(d) i–v	i **quaternary** · ii **fire** · iii **nebula** · iv **fold mountains** · v **subsistence farming**
(e) map	(viii) **1 cm = 0.5 km**; others depend on the 78M/11 sheet (map)

LAST-DAY CHECKLIST

tick when you can do it without looking (most repeated in 2023 / 2025)

- ☐ ☐ Big Bang (Lemaître, 13.8 bya, expansion = wider galaxy gaps, red shift) · Solar Nebula (nebula → disc → accretion → planets, 4.6 bya).
- ☐ ☐ Layers of Earth; lithosphere on asthenosphere; Wegener, Pangaea, Laurasia/Gondwana; Himalayas = India + Eurasia, fold mountains, marine fossils; GTS order: Precambrian → Palaeozoic → Mesozoic → Cenozoic (events).
- ☐ ☐ Wind: deflation, abrasion; suspension / saltation (1.5 m) / creep; mushroom rock, dune. Karst: solution; erosional vs depositional (stalactite, stalagmite, pillar).
- ☐ ☐ Bhutan climate: altitude, mountains (windward/leeward), monsoon; zones; alpine reasons; Lunana house.
- ☐ ☐ Time: 4 min = 1°, Bhutan GMT+6, east ahead; practise Sydney, Singapore, London, 155°W questions.
- ☐ ☐ Maps: IAC number (78E/11), RF → km per cm (÷ 100,000), 6-figure grid reference (right then up), contours, thread method, drainage patterns, plane table tripod.
- ☐ ☐ Outline maps: Arabian Desert, Urals, Hwang Ho, Black Sea, Tibetan Plateau, Caucasus, Yangtze, Yellow Sea; Drangme Chhu, Dochula, Jhomolhari, Simtokha, Phobjikha, Punakha, Wang Chhu, Pele La, Masa Gang, Tsenda Gang, Royal Manas.
- ☐ ☐ Population: growth rate calc, falling rate causes/effects, Gungtong, migration types, Central Place Theory (+ criticisms).
- ☐ ☐ Agriculture: subsistence, fragmentation → cooperatives, pests (IPM), climate-smart agriculture, self-sufficiency challenges.
- ☐ ☐ Minerals tree; 4 industry sectors; mining impacts; smart factory; industry location factors.
- ☐ ☐ Biodiversity levels; hotspot; SDF; vehicles; protection plan.
- ☐ ☐ Disasters: DM cycle, earthquake (Drop-Cover-Hold), GLOF (1994 Punakha), forest fire methods, EiE.
- ☐ ☐ In answers: count the points asked (TWO / FOUR) — one fact + one reason each; for “agree?” give your stand first.